

Centralized E-Commerce Platform for laptop sales and Auction Management System



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1. Introduction

Our Centralized E-Commerce Platform for laptop sales and Auction Management System is a web application built to facilitate the listing, auctioning and selling of laptops and other related electronic peripherals. The primary clients of this system include the buyers of electronic products and the vendors who wish to sell and distribute their devices. The system supports multiple vendors, allowing both individuals and organizations to register and sell their products online.

Given the increase in digitization, it has become a necessity to own a laptop due to its portability and importance in work and study. However, most customers face difficulty in finding quality products and vendors struggle to reach customers. This platform aims to address this issue.

Customers can easily browse the catalogue of items and choose to buy their desired product through the system. Additionally, this platform features an auction system that allows sellers to list their limited edition or special products for bidding within the system, and the buyers can place their bids within a specified range.

Our system consists of 3 major roles. Customer, Seller and Admin. A customer may register as a seller if needed. An account-based system is used to monitor and track role-specific data. Moreover, each role has its own functions within the system. While the primary focus is on laptops, the platform also supports selling other computer peripherals such as mice, keyboards, headsets and other accessories.

1.1 Motivation of the project

With the increase of technological advancements, laptops and other relevant devices have become essential items. Our platform addresses this need by providing a dedicated centralized platform which allows both sellers of devices and buyers of these devices to connect and perform business transactions for buying laptops and related devices.

Our platform automates the buying and selling process of laptops with the seamless connection between the two relevant parties. The seamless connection between the business transactions is managed by automating the processing of inventory and order taking, payment and tracking. Furthermore, it manages access to the database of the clients for the vendors. This platform will reduce the errors and wastage a manual system could have by automating all workflows. This automates the process of marketing your own products as the platform assists in the distribution process of the product.

Additionally, this platform assists small vendors or individuals to enter the market and perform their transactions with ease thus lowering the entry barrier for E-commerce infrastructure. Moreover, automating product listing and order processing will be extremely beneficial to vendors which saves time and resources on their end.

Also, this system helps sellers to easily keep track of their clients as a database that contains all relevant details of the customer is used by this platform. Our platform provides the necessary tools required for handling orders, inventory and customer analytics. Moreover, this streamlines

the process and helps sellers distribute the correct product to the correct client without any misconceptions.

To summarize, the motivation behind building this platform is to automate the process from the vendors side regarding listing a product, taking the order of the product and handling the payment of the product. From the side of the client this platform makes it easier for them to browse the catalogue of devices from many vendors, compare them with their reviews and purchase the most suitable laptop for their needs. This platform also provides an auction system the clients can utilize to get the best deal on a product. A platform such as this would be extremely beneficial for customers who struggle with choosing a laptop from multiple vendors. On top of that, for vendors the system is beneficial to reach their customers easily.

1.2 Objectives of the project

The objectives of this system are categorized into mainly primary and secondary goals.

Primary goals –

- Allow customers to browse, compare and purchase laptops and other accessories with ease.
- Enable vendors (Individual/ organizations) to list, manage and sell laptops and other computer peripherals through a centralized platform.
- Provide an auction feature for sellers to auction limited-edition or special products.
- Maintain an account-based system that supports three user roles such as Customer, Seller and Admin with role specific functionalities.
- Enable user authentication through email credentials and third-party identity providers such as Google, Facebook and other commonly used platforms.

Secondary goals –

- Support multiple vendors to ensure equal opportunities for all vendors.
- Provide seamless and user-friendly product distribution and inventory management system to ease the functionalities of the vendor.
- Maintain a database to store customer transactions.
- Allow customers to upgrade to sellers through the same account through role switching.
- Offer a wide variety of computer peripherals in addition to laptops.
- Ensure fairness and transparency in the auction process.

1.3 Scope

The system will enable users to browse, compare and purchase laptops and other related accessories, as well as allowing vendors to list and manage products.

The main features covered by our system are as follows:

- User authentication
- Role management: Customer, Seller, Admin

- Product listing and inventory management
- Auction functionality
- Order and payment handling
- Transaction and customer database

This platform targets multiple user segments.

1. Primary users

- University students
- IT professionals
- Business professionals

Our platform provides a quick efficient way for primary users to purchase a laptop without any hassle.

2. Secondary users

- Non-tech savvy individuals who may face difficulties in choosing the right product.

Our platform provides an efficient, easy method for secondary users to select a suitable laptop that aligns with their specific needs without any hassle.

This system will be a web-based application, accessible via modern browsers.

2. Analysis

2.1 Requirement gathering methods

To ensure that the proposed system is designed and implemented without any errors, our team analyzed existing systems in the current industry to identify its workflow and inefficiencies which could be improved in our proposed system. The following are the methods we collected requirements.

2.1.1 Observation

The primary method used to gather user requirements was through observing currently available websites that sell laptops and other electronic devices online. We observed the inner workings of websites such as Nanotek, Chama Computers, Laptop.lk and Laptopcare. In this requirement method our team observed the walkthrough of the end-to-end user journey on these platforms from product discovery to final checkout phase.

2.1.2 Interviews

To gain a much deeper understanding of the user's perspective, the team conducted brainstorming sessions (Internal interviews). In here the team members role-played as clients and discussed their personal experiences with existing systems when it comes to the purchasing of a laptop or any other electronic product.

2.2 Similar system identification

Our team observed and followed through the workflows of similar systems. One main fact we identified is that, with the features we are planning on implementing, there are currently no existing systems for our proposed system. There are similar systems which engage in the selling of Laptops and electronic products. However, our proposed system plans to implement an Auction feature to our products. Furthermore, we are planning to implement the system as a multi-vendor system where buyers can register within the system and list their products for sale. These combined features are unfound in the current similar systems we observed.

We mainly observed the following systems,

- Nanotek
- Chama Computers
- Laptop.lk
- Laptopcare

2.2.1 Nanotek

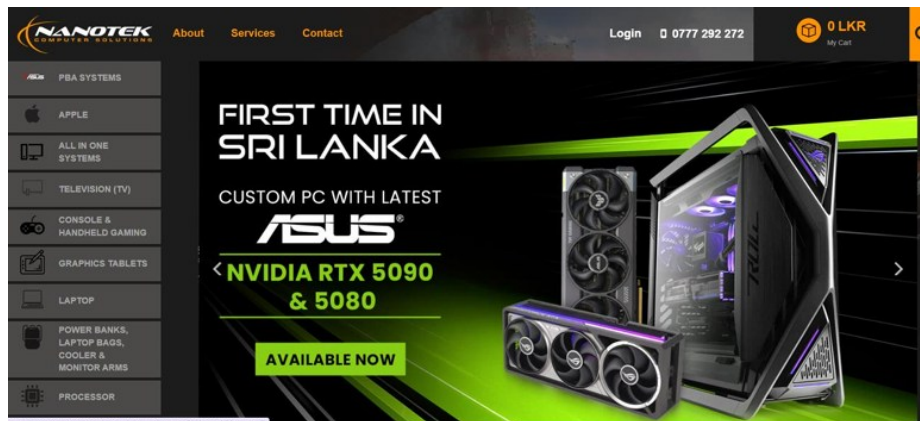


Figure 1 - Nanotek similar system

Nanotek is a Sri Lankan online retail system which allows customers to browse through it and purchase computer hardware, gaming equipment, Laptops and other IT related products through its web-based platform. This system allows the users to browse through products by categories, view detailed product information, and place orders online. This website follows a structured front-end design with clear navigation menus, product listings and user interaction elements such as buttons and forms.

A. Key features

- Comprehensive product catalog with categorization to include custom PC systems, Laptops, Apple products and specialized hardware such as High-end GPUs.

- Allow customers to search for laptops based on advanced filtering through specifications such as CPU mode, GPU, screen size, price range and brand.
- Offers user to either pay online or to request a quotation to be sent to their email inbox.
- Allows customers to add multiple items to a cart and generate a total bill.
- Displays detailed technical specifications of a product.

B. Strengths

- Labels items as “In Stock”, “Out of Stock” and “Pre-order” to ensure availability and removes “Add to cart” in unavailable products.
- Provides specific sections for warranty assurance, custom orders and home delivery with clear terms and conditions.
- A navigation bar on left and header search bar provides multiple methods for users to search for products.

C. Weaknesses

- Slower response time for inquiries as inquiry portal relies on traditional communication methods for customer support.
- To purchase a product a user must log in first.
- Lack of customer reviews or community comments.

D. Gap Analysis and System Motivation

- While existing systems use traditional bank gateways, our transition to Supabase and Vercel (with PayHere integration) ensures faster, more responsive UI with real-time updates for inventory and dynamic pricing.
- Our system aims to categorize products based on User’s usage (Business, Consumer, Gaming) which can be beneficial to non-technical users to find products easily.

2.2.2 Chama Computers

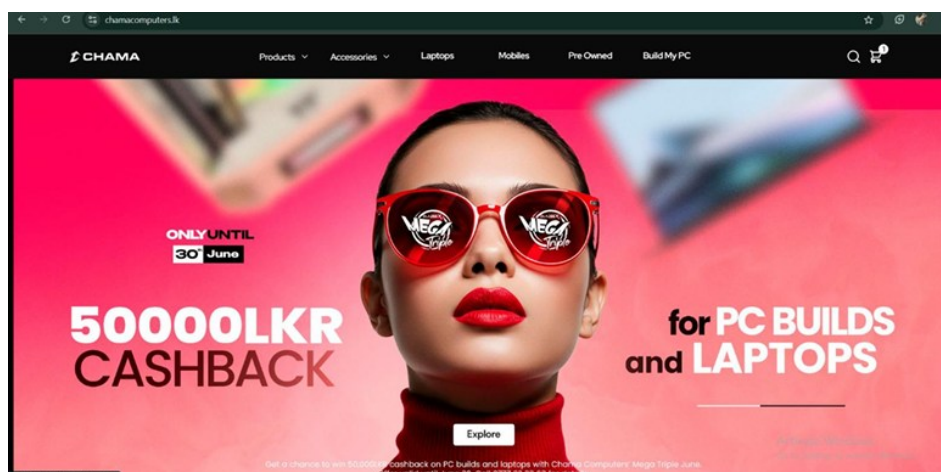


Figure 2 - Chama computers similar system

Chama computers is a Sri Lankan IT retail and services website which demonstrates a wide range of computer hardware, accessories and custom PC builds which allows customers to browse through the best-selling products and explore items which are categorized efficiently. This platform provides a structured navigation with product listings, special offers and customer engagement features providing an organized frontend interface for users to interact with the system.

A. Key features

- A specialized “Build My PC” module which allows customers to select individual components they need to create a custom Desktop.
- A diverse inventory which includes standard hardware, Pre-owned and mobile categories.
- Offers multitude of promotions from Cashback promotions to Special offers for high-demanding products.

B. Strengths

- Highlights “Save RS. X” directly on product to drive customers to make immediate purchase decisions.
- Detailed technical specifications of the products.
- Add to cart and buy now options for available inventory.

C. Weaknesses

- Vast number of categories confuse casual buyers and make them feel overwhelmed.
- Features high-quality product images with no image optimizations which leads to slow load times (High latency)
- Larger hero banners and oversized images occupy too much space in home page, forcing users to scroll excessively to reach actual product listings.

D. Gap Analysis and System Motivation

- More user friendly product categories to not make the casual buyer feel overwhelmed.
- Performance optimization to solve speed issues.
- A compact UI design to minimize scroll-heavy nature of the system.

2.2.3 Laptop.lk

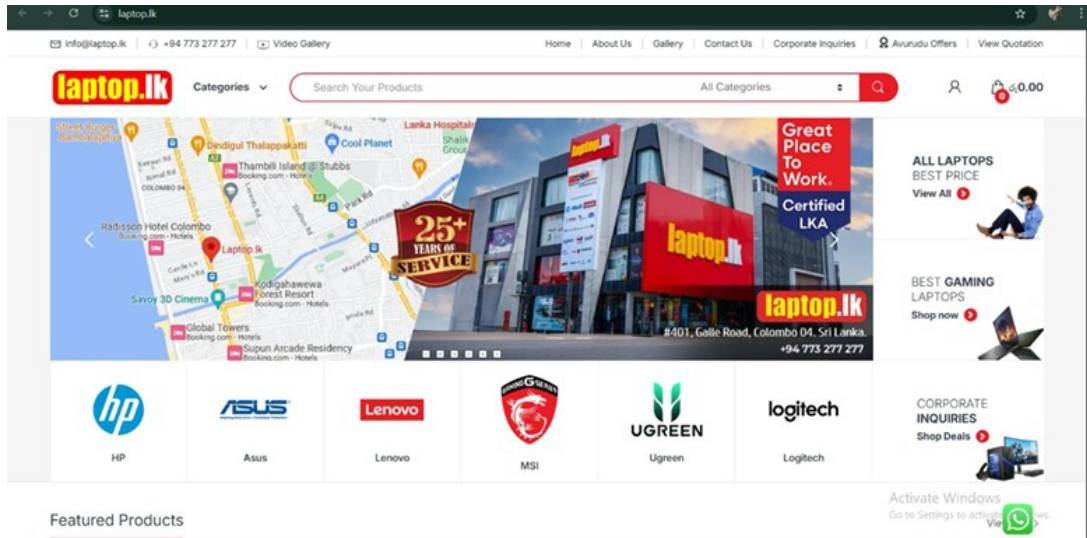


Figure 3 - Laptop.lk similar system

Laptop.lk is a Sri Lankan IT retail e-commerce platform which provides a comprehensive range of laptops, desktops, printers, accessories and computer components from leading global brands. The site provides the users with a properly structured navigation through the product categories which enables users to explore different products and access detailed information on each product.

A. Key features

- Offers standard immediate purchase and “Add to Quote” options.
- Advanced filtering system for users to filter hardware by technical specs, price, brand, processor, RAM, and GPU.
- Categorizes products into Commercial, Consumer and Gaming.
- Customer support through WhatsApp and Direct Calling.
- Guest account can purchase a product and checkout without making an account.
- Multiple payment methods such as Bank transfers, Visa, American Express and Koko payment.

B. Strengths

- Detailed documentation and specifications about products with review tabs for each product.
- Promotional offers during festive seasons.
- Building trust through long standing industry presence (25+ years of service).

C. Weaknesses

- Even though support through WhatsApp is provided, there is a lack of instant technical troubleshooting.
- Traditional UI/UX.

- Quotation system requires users to fill out their details manually which slows down the B2B process.

D. Gap Analysis and System Motivation

- While Laptop.lk focuses on static quotations, our system introduces an Auction module which users can acquire high-demand products at competitive prices.
- Faster and reliable experience through Supabase and Vercel than traditional E-commerce platforms.
- Refined categorization that suits non-technical users.

2.2.4 Comparative Summary of Systems

Feature	Nanotek	Laptop.lk	Chama computers	LaptopHub (Our proposed system)
Core Tech stack	Traditional web stack	Traditional web stack	Traditional web stack	Supabase + Next JS
Categorization	Component based	Brand/ Usage based	Vast/ Complex	User intent (Business, Consumer, Gaming)
Auction module	Not available	Not available	Not available	Available
Image optimization	Standard	Standard	Unoptimized	Optimized
UI/UX	Moderate	Traditional	Excessive scroll	Compact and responsive

Table 1 - Comparative summary of systems

2.3 Feasibility study

A feasibility study was conducted to analyze whether our proposed system was a feasible project.

- **Technical Feasibility:**

- The web-based system can definitely be built using current technology as we have chosen a solid technology stack to build it upon. The technology stack includes Supabase for the backend as the database and V0 by Vercel for building the frontend (user interface) of the system. Furthermore, the system will be responsive, so that the user can access and view it from a PC, tablet, or mobile device.

- **Operational Feasibility:**

- By providing a single digital market for laptops, the Marketplace will be able to create a simpler and less complicated method to buy and sell these products. The Vendors will have an easier time managing their individual businesses using a common inventory and order tracking system and will provide customers with an easy method to locate, compare and submit bids on the products they desire. The Organization of creating roles within the Marketplace (Customer, Seller, and Admin) provides for a more efficient and orderly way to manage the Marketplace Member.
- **Economic Feasibility:**
 - The project offers long term economic benefits by lowering the entry barrier for small time vendors and individuals to enter the e-commerce market. By automating workflows such as product listing and payment handling, the system reduces the manual errors and resource wastage associated with traditional sales methods.
- **Legal Feasibility:**
 - By using necessary authentication methods, the system will remain compliant with regulations to protect user data privacy. Role-based access control (RBAC) enables access to customer's sensitive records and transaction information to be limited to those who have been granted permission, including vendors and administrators.

2.4 Requirement analysis

Through the requirement gathering phase we have identified key functionality that should be covered by our proposed system.

2.4.1 Functional Requirements

A. User Registration and Login

- Users must be able to login to the system through their emails or third-party accounts (E.g. Google, Facebook)
- Users must be able to register and login according to their usage. Seller must be able to login as a seller and a buyer must be able to login as a buyer.

B. Role-based Access control

- System must support the main three roles of the platform with relevant access and permissions.
- A seller must be able to add new products and maintain them.
- A buyer must be able to browse product catalogue and purchase what they want.
- An admin must be able to maintain the internal work of the platform seamlessly.

C. Product listing and inventory management

- Sellers should be able to add, update and remove products.
- Inventory quantity should be tracked and updated with each purchase.

D. Product browsing and comparison

- Customers must be able to browse and search for products.
- Customers must be able to purchase the product they need through the platform.
- E. Auction management
 - Sellers can list items for auction.
 - Buyers can place bids within the auction period.
 - Winning bid is automatically handled after auction time expiry.
- F. Order placement and payment handling
 - Customers can place orders and complete payments.
 - Vendors get notified about relevant orders.
- G. Admin management
 - Admins should be able to manage users, moderate content and overall monitor the system.
 - Any issues that arise within the platform must be handled by an admin.
- H. Review system
 - Customers should be able to rate and review products or vendors depending on their service.

2.4.2 Non-functional Requirements

- A. Scalability
 - As the number of vendors increases, the system shall utilize its cloud-based architecture to scale horizontally and vertically.
- B. Performance
 - System should respond to user actions within 5 seconds.
 - During a user login, the information should be verified within 5 seconds.
 - System should support at least 100 concurrent users without any performance degradation.
- C. Security
 - Our platform implements role-based access control (RBAC). The main user roles include: Admin, Seller, Customer, Guest (Limited read-only access without purchase or bidding ability).
 - All users require authentication to access protected features.
 - Passwords are securely stored using industry-standard encryption techniques.
 - User sessions are managed using secure HTTP-only cookies with automatic token refresh to maintain security.
 - Each subsystem should be accessible based on the user's role.
 - Admin functions should be only available to admin users.
 - Sellers should be able to only modify their own products and customers should only be able to access their own order history.
- D. Availability
 - The system should be operational 99.9% of the time during business hours.
 - Total system downtime should not exceed 45 minutes per month.
 - Scheduled maintenance must be conducted during off-peak hours.

E. Usability

- The User interface should be user friendly even for non-tech users.
- A new user should be able to complete a purchase or place a bid within a short amount of time.

F. Maintainability

- Codebase and architecture should allow for easier updates and fixes.
- Use of GitHub-integrated auto-deploys to ensure that bug fixes can be pushed to production seamlessly without system downtime.

G. Responsiveness

- The application must work on multiple screen sizes (PC, Tablet and mobile)

2.4.3 Hardware Requirements

Hardware component	Recommendation
Server type	Cloud based
CPU	Minimum 4 core processor
RAM	Minimum 4 GB
Storage	SSD with 100 GB minimum
Bandwidth	Sufficient to handle concurrent users
Backup	Daily backup system (Cloud/ External)

Table 2 - Hardware requirements of proposed system

Note: The current system is being developed and deployed using Supabase Free Tier cloud infrastructure. Hardware resources are mainly managed by the service provider. If system usage increases significantly, higher-tier cloud resources or dedicated infrastructure may be adopted.

2.4.4 Software Requirements

Category	Component implementation
Database	Supabase
Authentication	Supabase authentication
Payment integration	PayHere sandbox
File Storage	Supabase storage
Hosting	Vercel
CI/CD	GitHub-integrated (auto-deploys)
Version control	GitHub

Table 3 - Software requirements of proposed system

Note: In the initial project proposal, it was stated that the system was intended to use the following,

- Database - PostgreSQL / MySQL/ Firebase Firestore
- Authentication - OAuth 2.0 + Email Auth
- Payment integration – Bank gateway
- File storage – Firebase storage

However, after requirement analysis phase, the team transitioned to a Supabase integrated architecture to ensure that the database, authentication and file storage to be used in a single

ecosystem. Additionally, PayHere was selected to implement payment integration to the system to better suit the local market requirements.

3. Design

3.1 Design Diagrams

3.1.1 UML Use Case Diagram

The following UML Use Case Diagram efficiently depicts the interactions that would take place between the main actors and the proposed system. Customers are the end users who register and log in to the system. They can browse and compare products and order them or place a bid on an auctioned product. Furthermore, they can review a product and even upgrade their account to a seller status. Seller is responsible for creating and managing auctions and listing and selling products. They handle the inventory, customer orders and manage revenue. The Payment Gateway handles secure payment and Admin oversees all system level tasks within the system.

- **Customer**
 - Register and Login
 - Browse Products
 - Compare products
 - Place Order
 - Place Bid on Auction
 - Rate and Review
 - Upgrade to seller
 - View Auction Status
 - Add items
- **Seller**
 - Register and Login -
 - View Auction Status -
 - Create Auction
 - Sell Products
 - Manage Inventory
 - List Products
 - View Orders and Customer Details
 - Handle Order Fulfillment
 - Collect Revenue
- **Payment Gateway**
 - Process Payment
- **Admin**
 - Moderate Content
 - Verify User
 - Monitor System
 - Manage Database

- Manage Users
- Prepare Reports

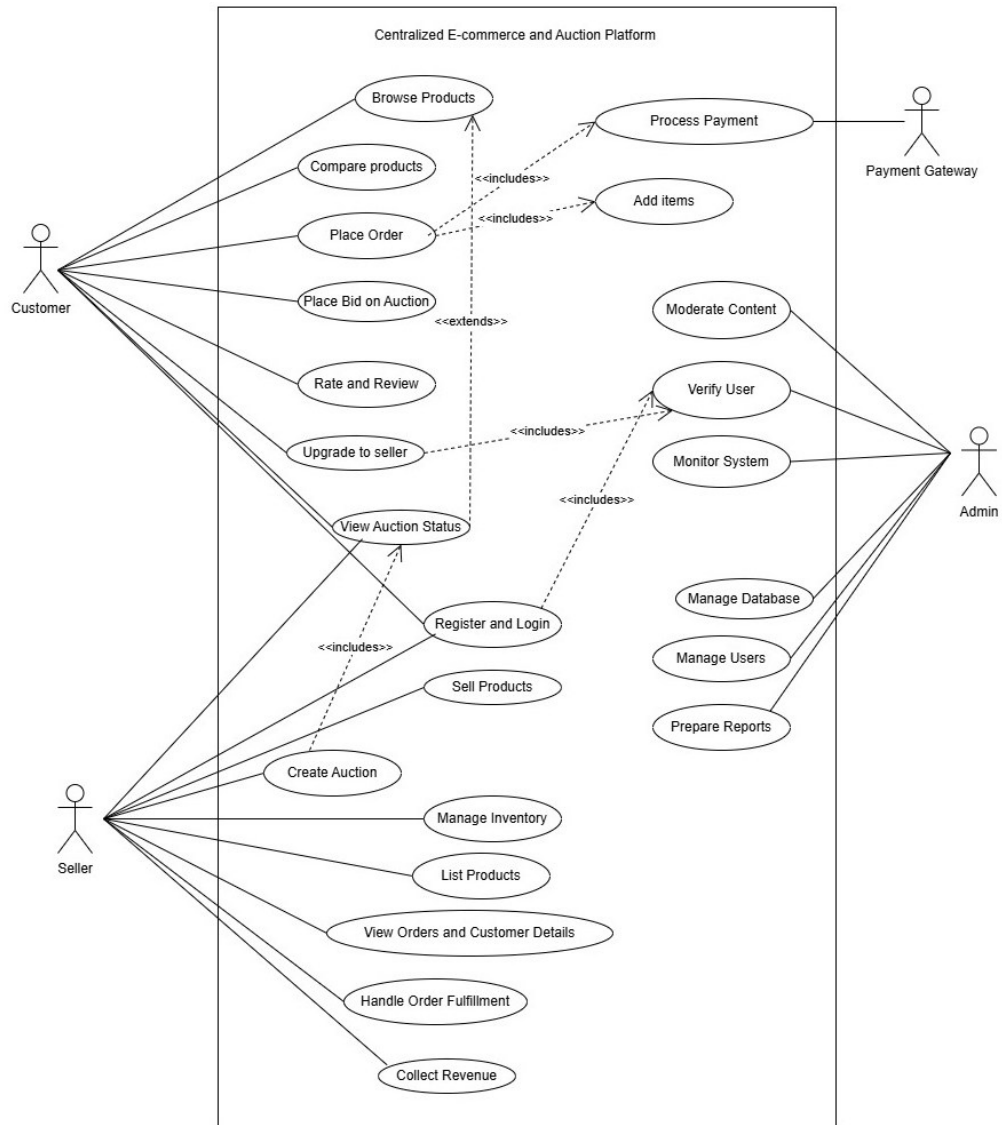


Figure 4 - UML Use case diagram of the proposed system

3.1.2 UML Class Diagram

The following UML Class diagram illustrates the structure of our proposed system. It shows the main classes, their attributes, methods and relationships between each of them. It defines how core entities are organized and how they interact within the system.

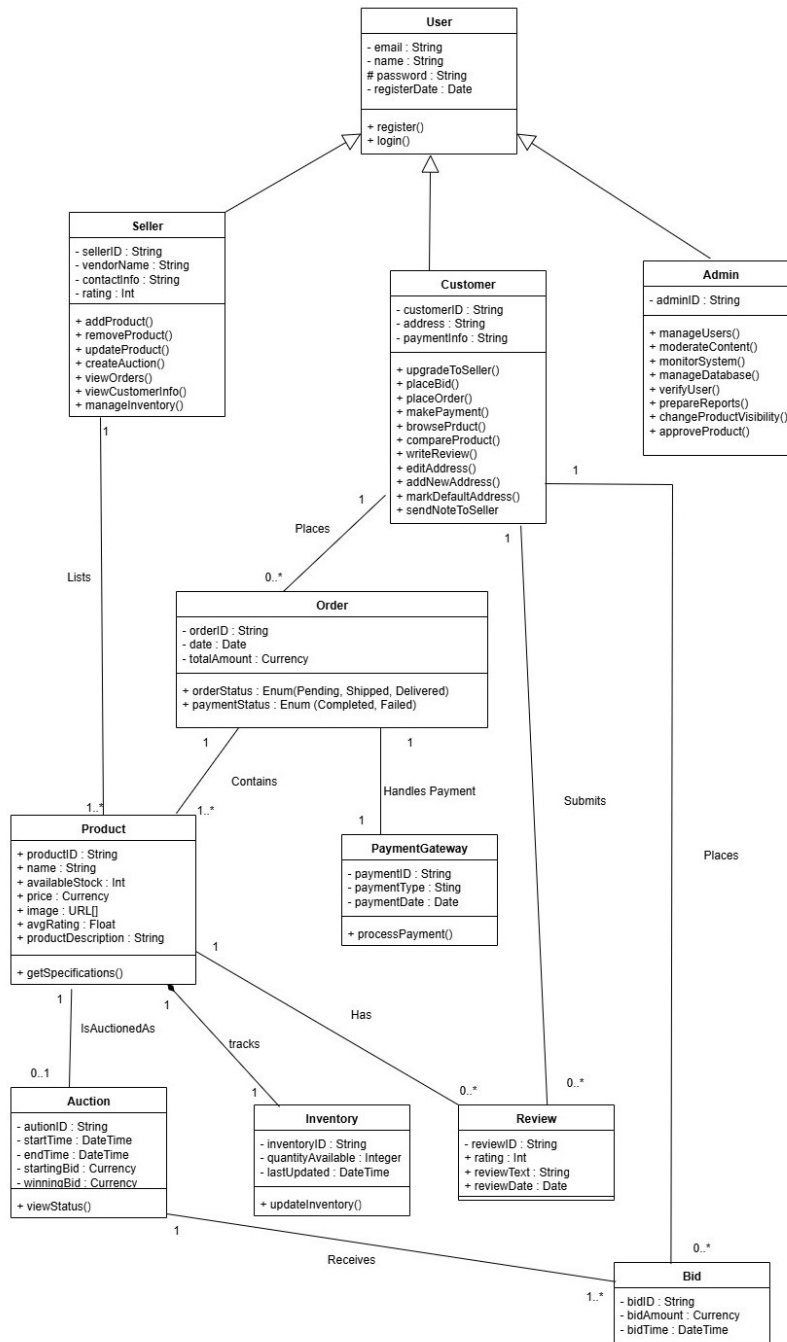


Figure 5 - UML class diagram of the proposed system

3.1.3 Entity Relationship Diagram (ERD)

The Entity Relationship Diagram of our system consists of all the entities which interact with our system. This diagram accurately represents how all the 10 entities interact with each other.

- **USERS table**
 - id (Primary Key)
 - name
 - role
 - rate
 - created_at
 - updated_at
- **ADDRESSES table**
 - id (Primary Key)
 - user_id (Foreign Key, references USERS.id)
 - street_line_1
 - street_line_2
 - city
 - state
 - postal_code
 - country
 - is_default
- **PRODUCTS table**
 - id (Primary Key)
 - seller_id (Foreign Key, references USERS.id)
 - name
 - brand
 - price
 - specs
 - images
 - badge
 - stock
- **WISHLISTS table**
 - id (Primary Key)
 - user_id (Foreign Key references USERS.id)
 - product_id (Foreign Key references PRODUCTS.id)
- **REVIEWS table**
 - id (Primary Key)
 - user_id (Foreign Key, references USERS.id)
 - product_id (Foreign Key, references PRODUCTS.id)
 - rating

- comment
- **AUCTIONS table**
 - id (Primary Key)
 - product_id (Foreign Key, references PRODUCTS.id)
 - seller_id (Foreign Key, references USERS.id)
 - starting_bid
 - reserve_price
 - start_time
 - end_time
 - status
- **BIDS table**
 - id (Primary Key)
 - user_id (Foreign Key, references USERS.id)
 - auction_id (Foreign Key, references AUCTIONS.id)
 - amount
 - created_at
- **ORDERS table**
 - id (Primary Key)
 - customer_id (Foreign Key, references USERS.id)
 - total_amount
 - status
 - payment_status
 - shipping_address
- **ORDER_ITEMS**
 - id (Primary Key)
 - product_id (Foreign Key, references PRODUCTS.id)
 - order_id (Foreign Key, references ORDERS.id)
 - quantity
 - unit_price
 - total_price
 - select_options
- **PAYMENTS**
 - id (Primary Key)
 - user_id (Foreign Key, references USERS.id)
 - order_id (Foreign Key, references ORDERS.id)
 - transaction_id
 - amount
 - provider
 - status

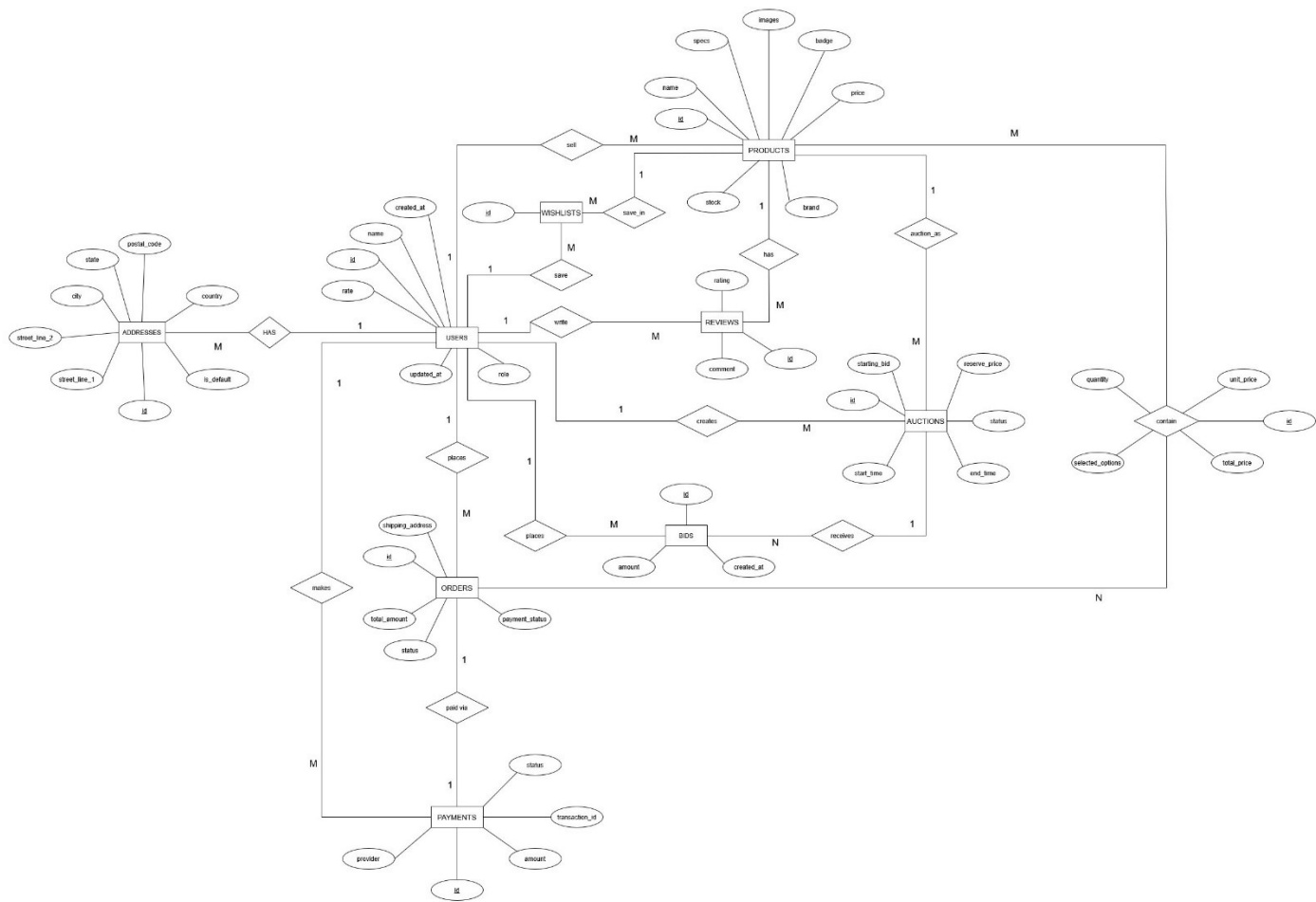


Figure 6 - ER diagram of the proposed system

3.1.4 UML Activity Diagram

The following UML diagrams accurately depict the workflow of the following modules.

- Product purchase

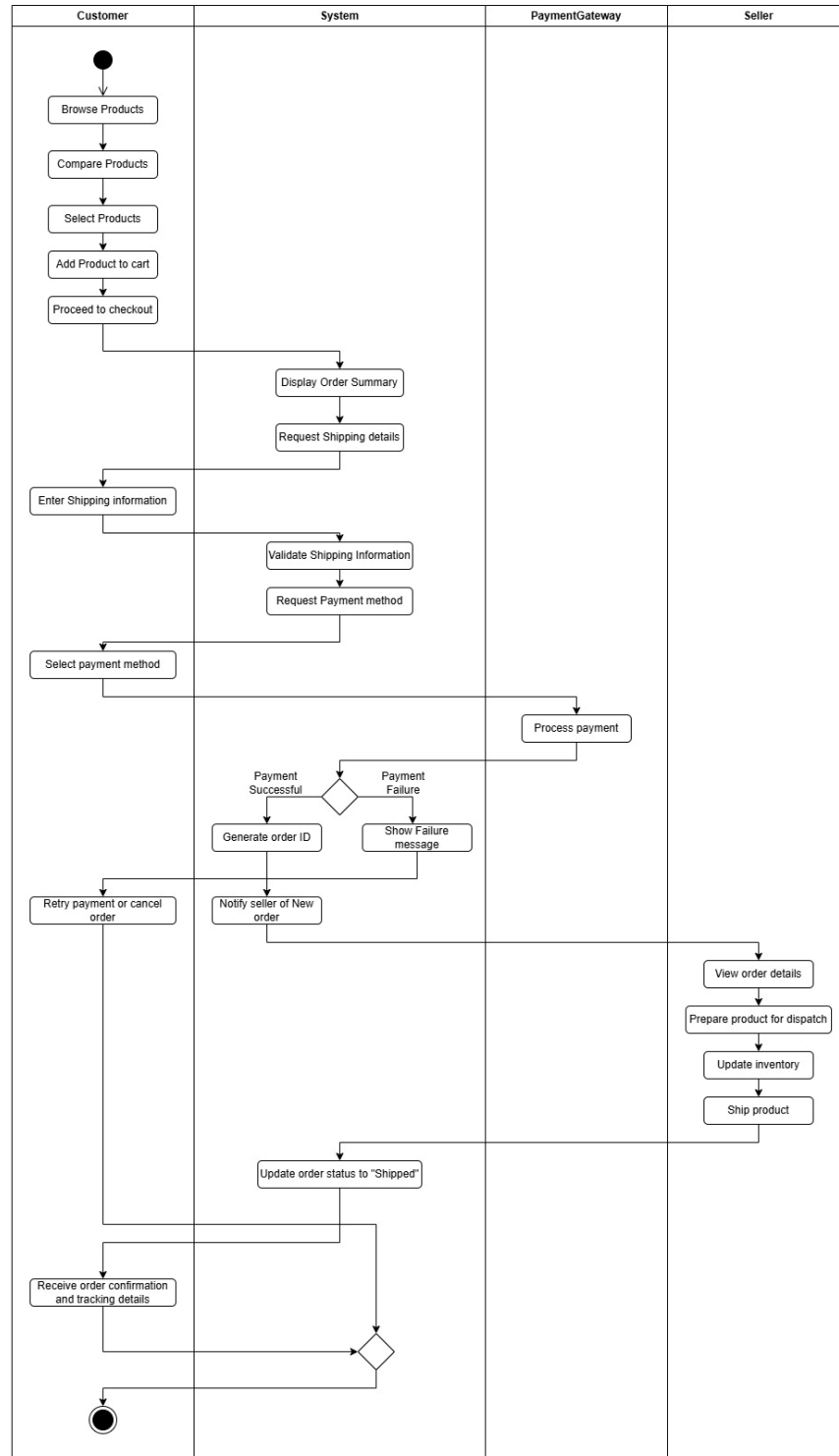


Figure 7 - UML activity diagram for product purchase module of the proposed system

- Auction and bidding

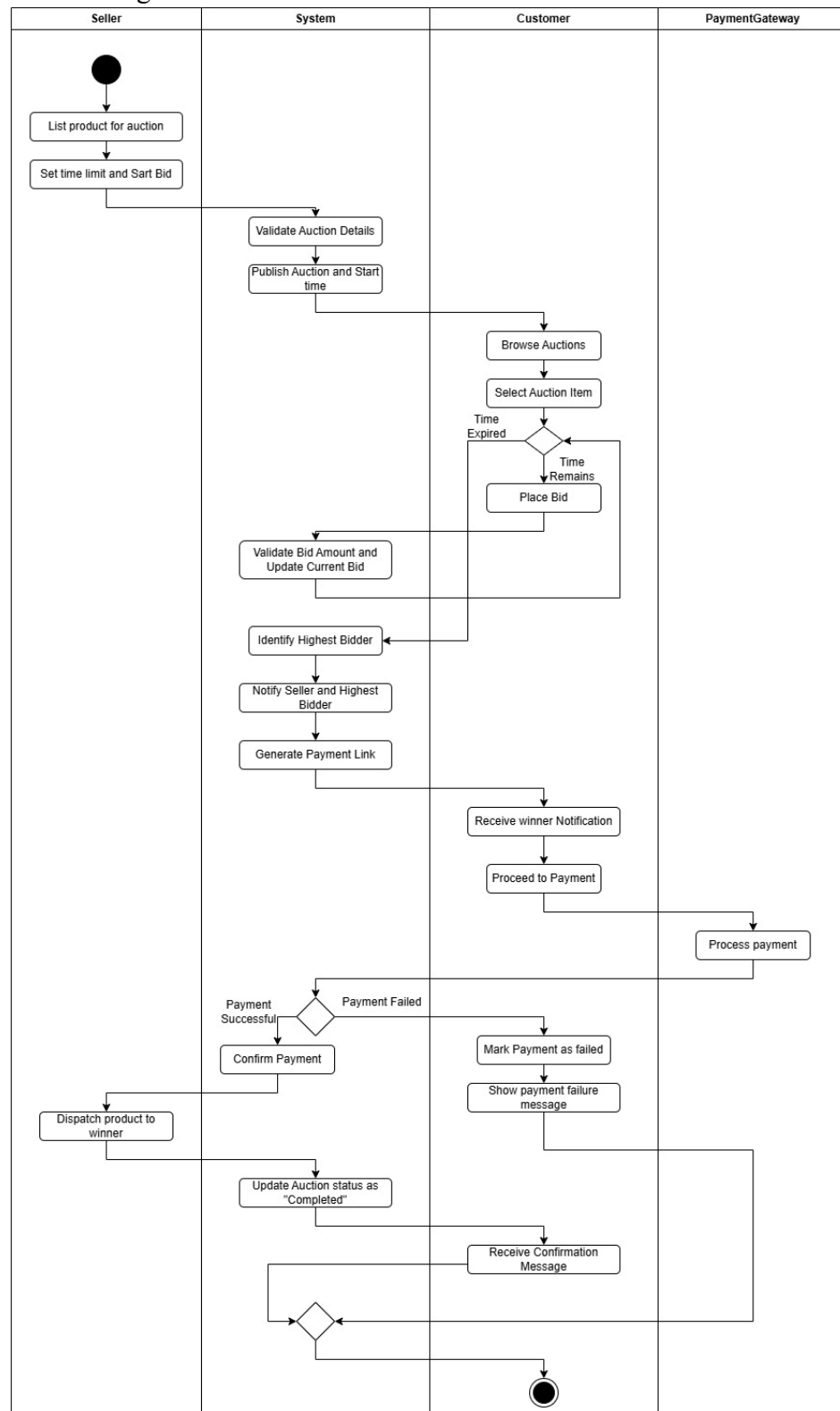


Figure 8 - UML activity diagram for auction and bidding module of the proposed system

- Sign up and Login

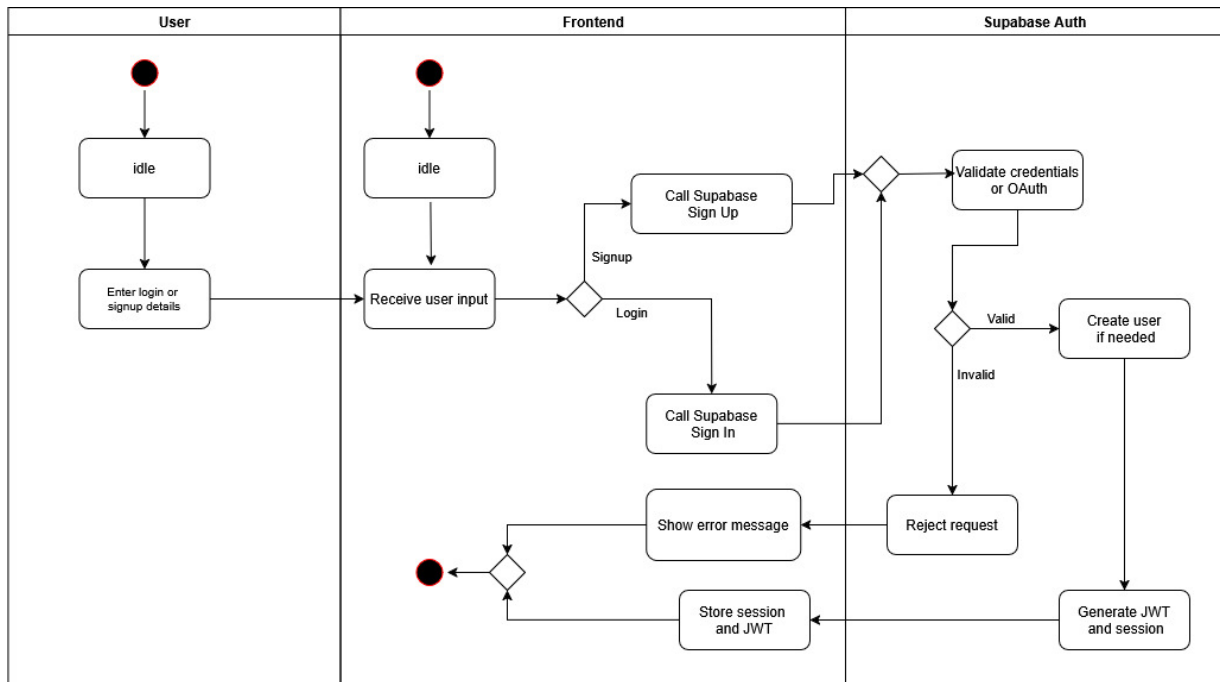


Figure 9 - UML activity diagram for login and sign up module of the proposed system

3.2 System Design

3.2.1 System Wireframe

- Home page
 - This is the common page every user visits. Users can browse products in this page.

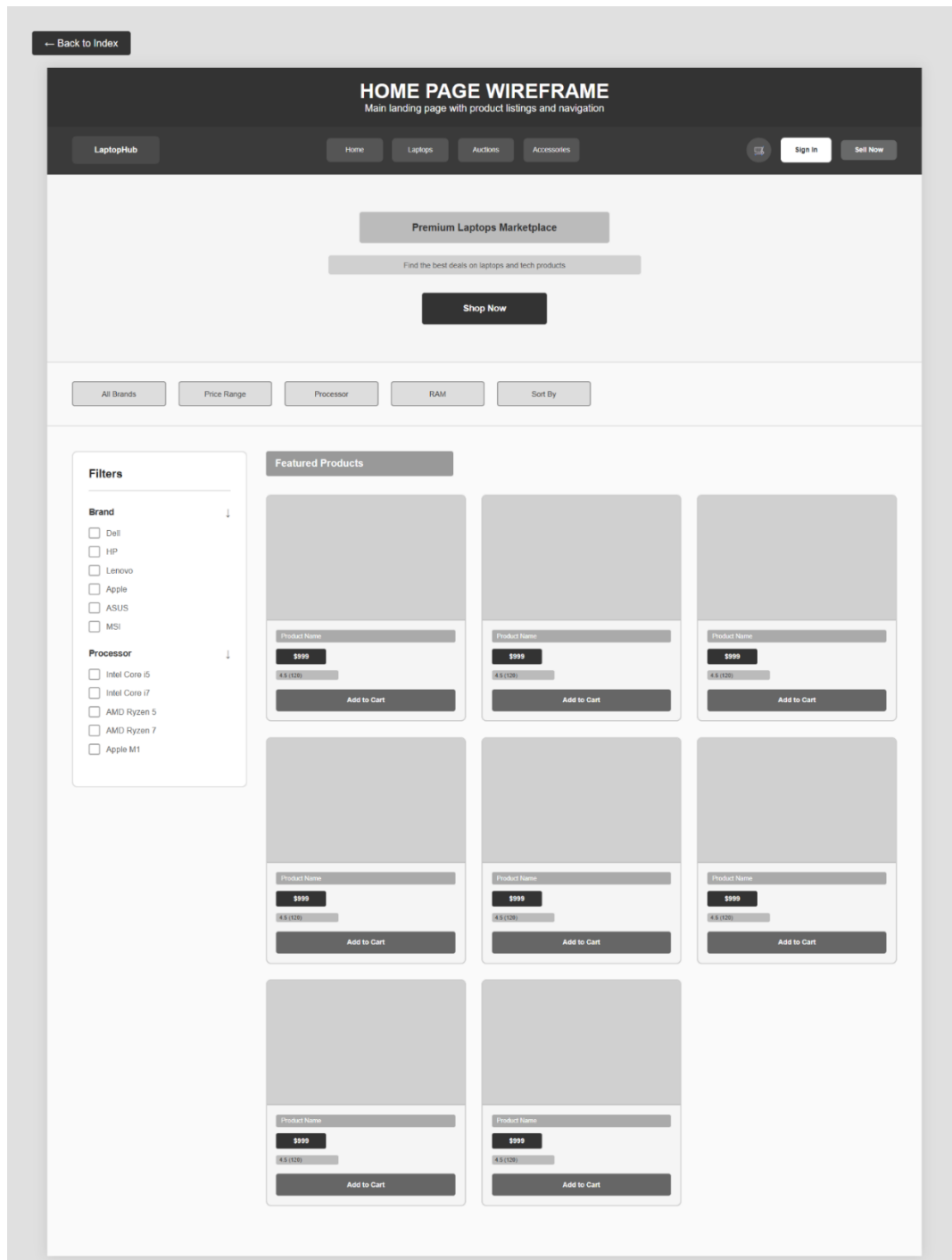


Figure 10 - Home page wireframe

- Register page (Sign up)
 - This page allows a new user to register and make an account.

The wireframe for the Register page is divided into two main sections. On the left is a dark sidebar containing the 'LaptopHub Logo' at the top, a large 'Laptop Image Background' placeholder in the center, and a list of three items at the bottom. The main content area on the right is light gray and titled 'REGISTER PAGE WIREFRAME' with the subtitle 'New user account creation interface'. It features a 'Create an account' button, followed by a prompt to 'Enter your details below to create your account'. The form includes fields for 'Sign as' (with 'Buyer' and 'Seller' options), 'Full Name', 'Email Address', 'Password', and 'Confirm Password'. Below these is a checkbox for 'I agree to the Terms of Service and Privacy Policy' and a 'Create Account' button. At the bottom of the main area are links for 'Or', 'Google', 'Facebook', 'Already have an account?', and 'Sign in'.

Figure 11 - Register page wireframe

- Login page
 - This page allows a registered user to log in to the system.

The wireframe for the Login page follows a similar layout to the Register page. The dark sidebar on the left includes the 'LaptopHub Logo', a 'Laptop Image Background' placeholder, and a quote: 'The best place to buy and sell premium laptops. Secure, fast, and reliable.' with a 'Learn more' link. The main content area is titled 'LOGIN PAGE WIREFRAME' with the subtitle 'User authentication and sign-in interface'. It starts with a 'Welcome back' button and a prompt to 'Enter your email to sign in to your account'. The form has fields for 'Email Address' and 'Password', a 'Remember me' checkbox, and a 'Forgot password?' link. A 'Sign in' button is positioned below the form. At the bottom, there are 'Or', 'Google', 'Facebook', 'Don't have an account?', and 'Sign up' links.

Figure 12 - Login page wireframe

- Account page
 - This page allows a user to view their account and manage account settings.

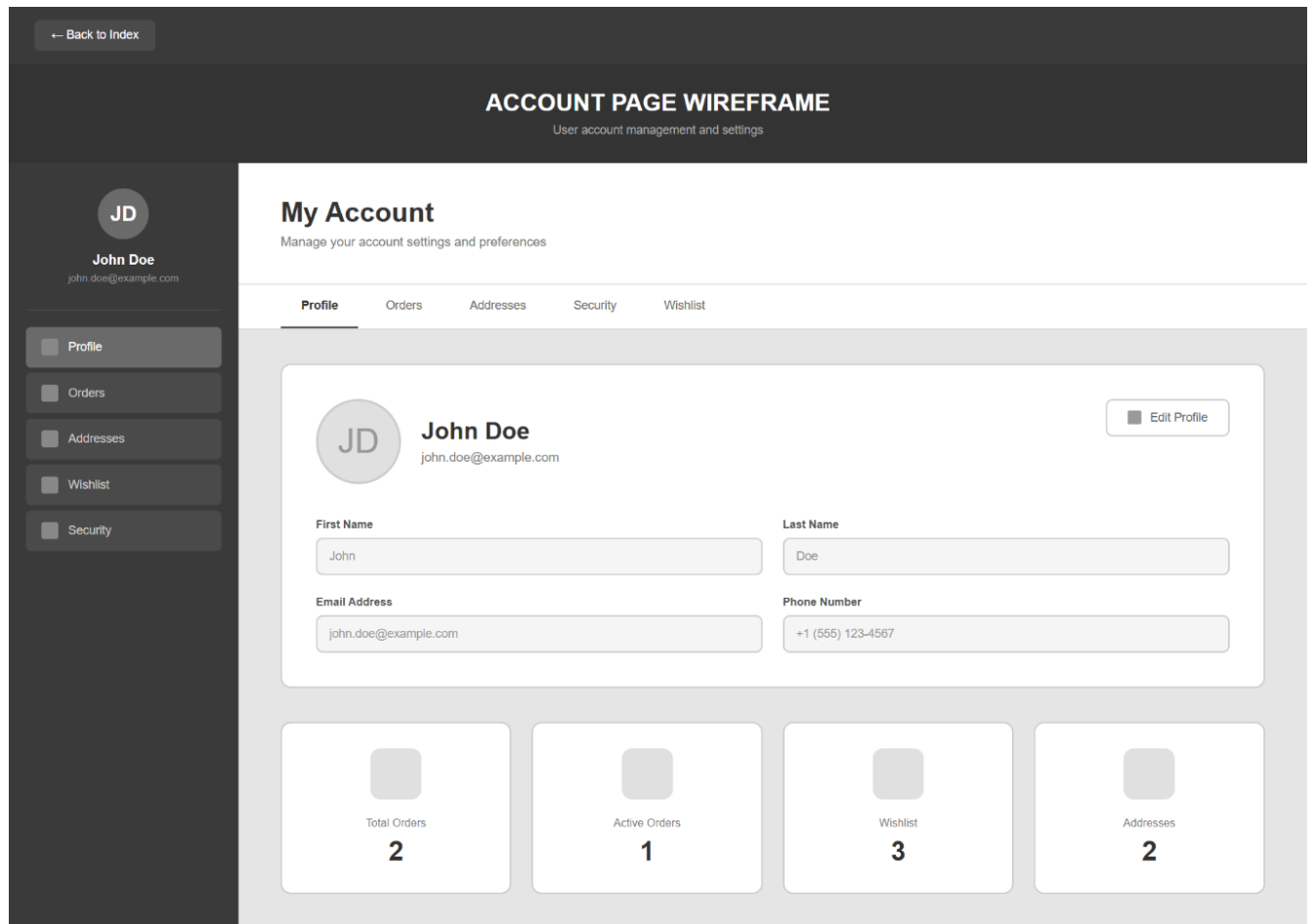


Figure 13 - Account page wireframe

- Product details page
 - This page allows a user to view all the product details of a product that they have selected.



Figure 14 - Product details page wireframe

- Auction page
 - This page allows a user to view all the current on-going auctions of the products.



Figure 15 - Auction page wireframe

- Auction details page
 - This page allows a user to view specific information regarding a product which is up for auction. They can place their bids on the product on this page.

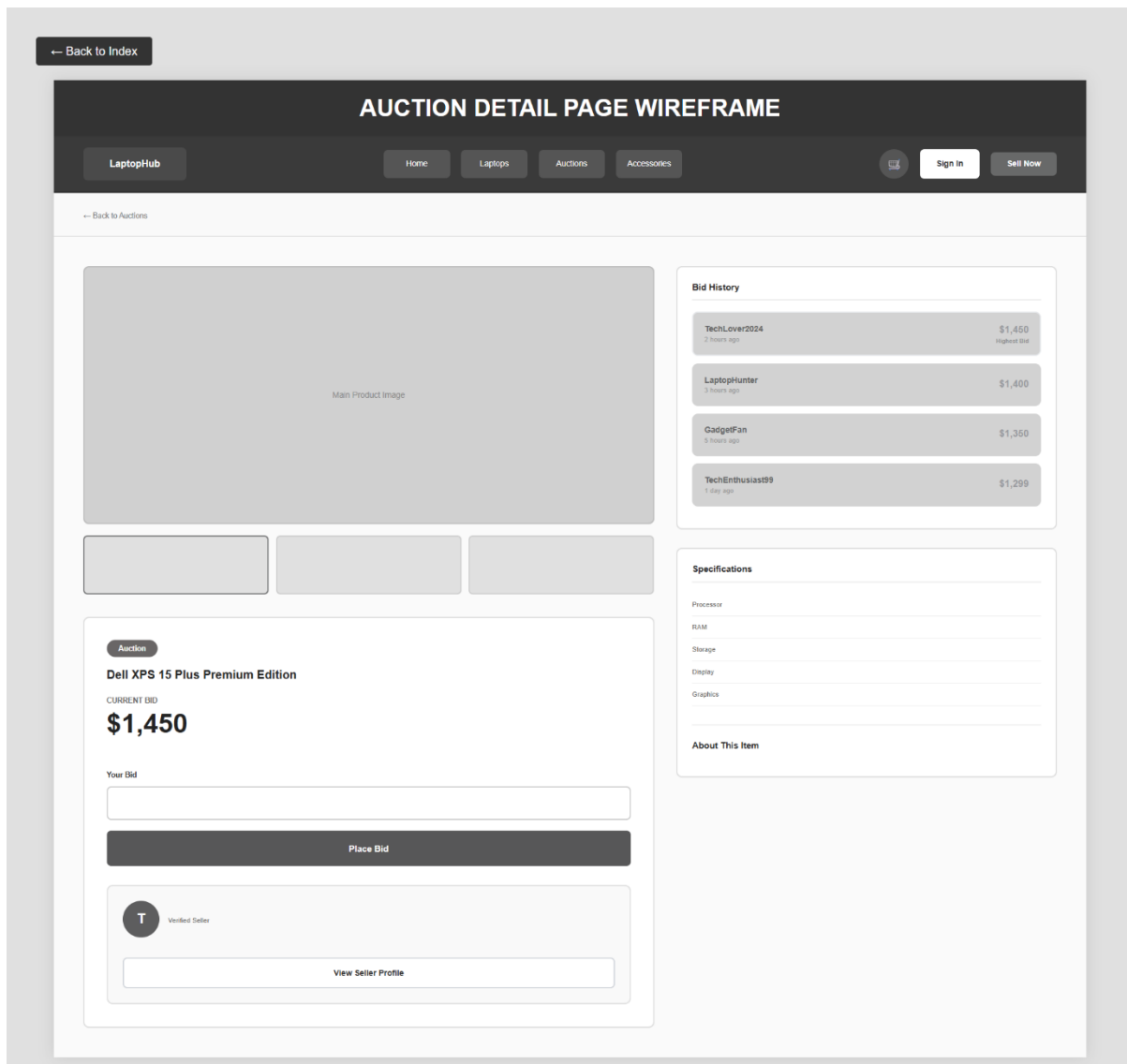


Figure 16 - Auction detail page wireframe

- Shopping cart page
 - This page allows a customer to view the products they have put into the shopping cart and go through with the payment.

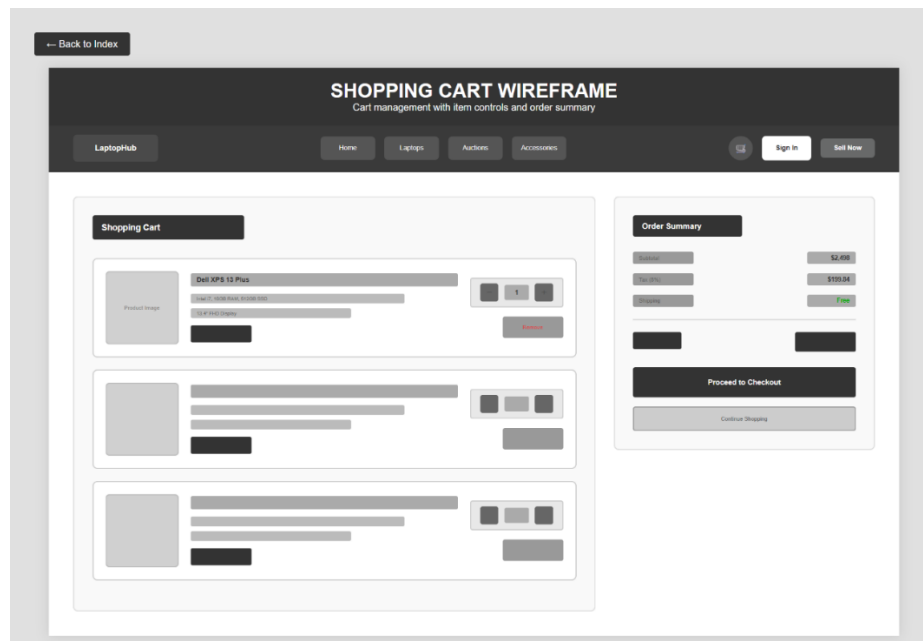


Figure 17 - Shopping cart wireframe

- Seller dashboard
 - This page allows a seller to view all sales overview, order management and performance metrics for the seller.

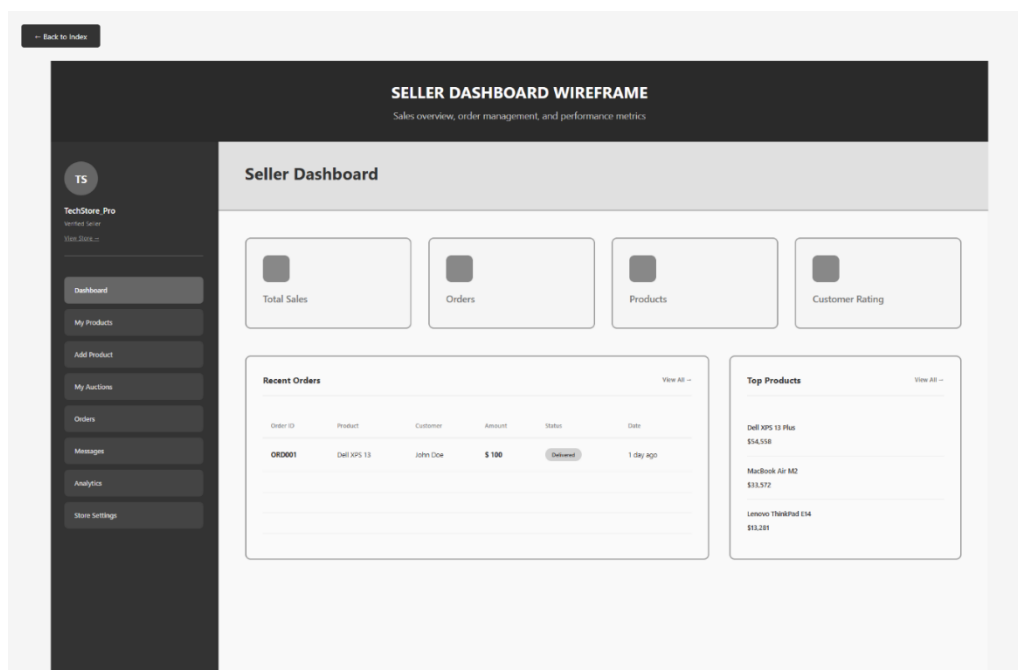


Figure 18 - Seller dashboard wireframe

- Admin-Seller page
 - This page allows a seller to manage and monitor their products and sales.

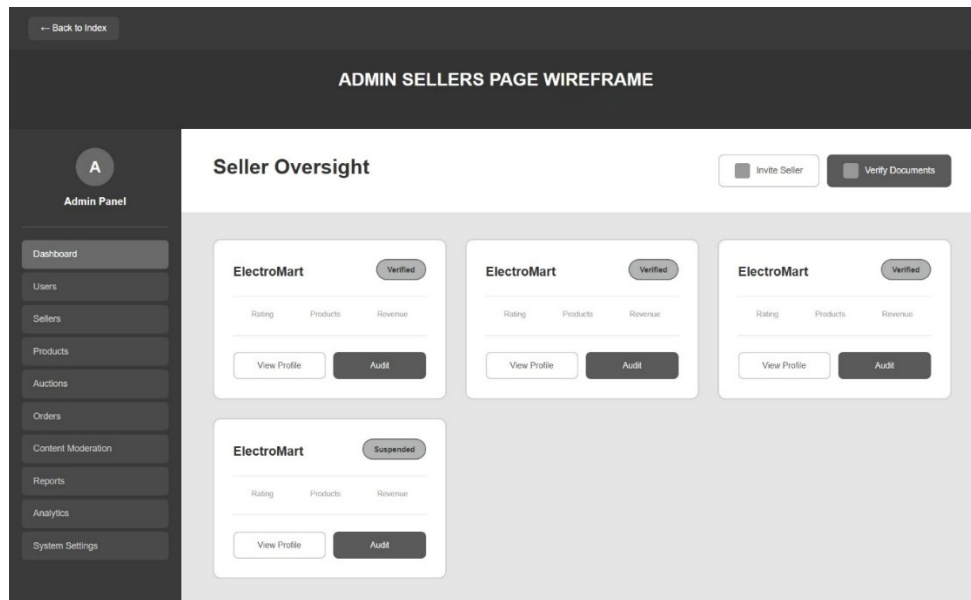


Figure 19 - Admin-seller page wireframe

- Admin dashboard
 - This page allows a system to monitor the system, manage the platform and control the users of the proposed system.

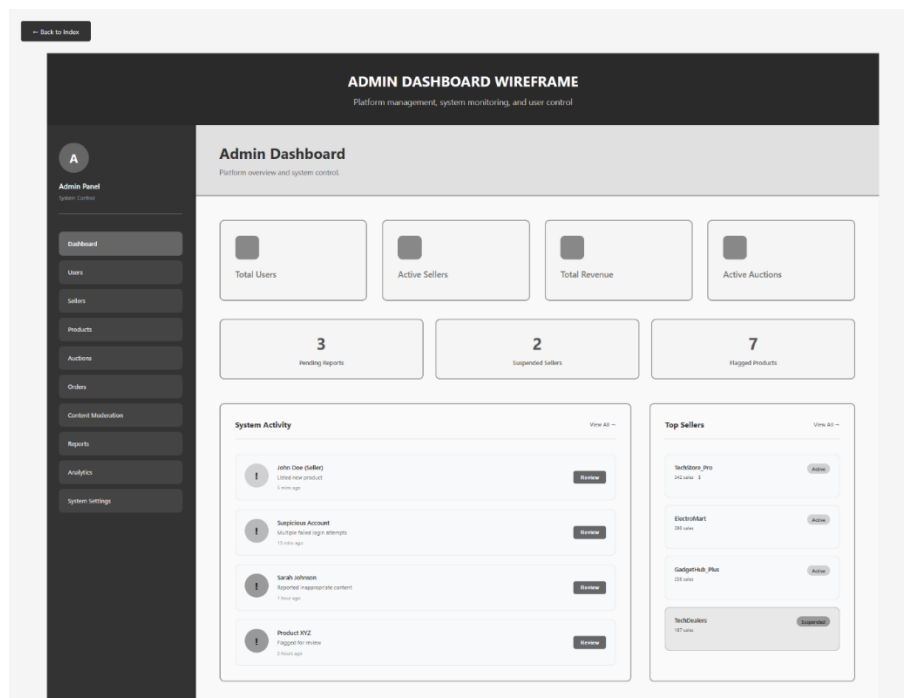


Figure 20 - Admin dashboard wireframe

- Footer
 - This contains all the information that will be displayed in the footer of each web page.

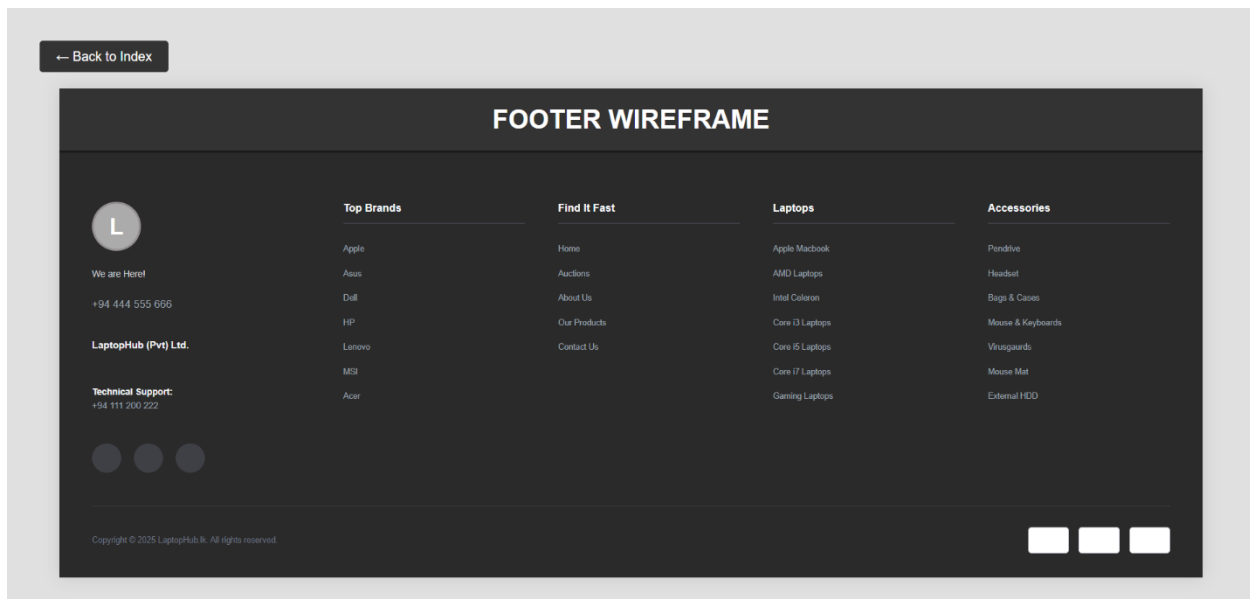
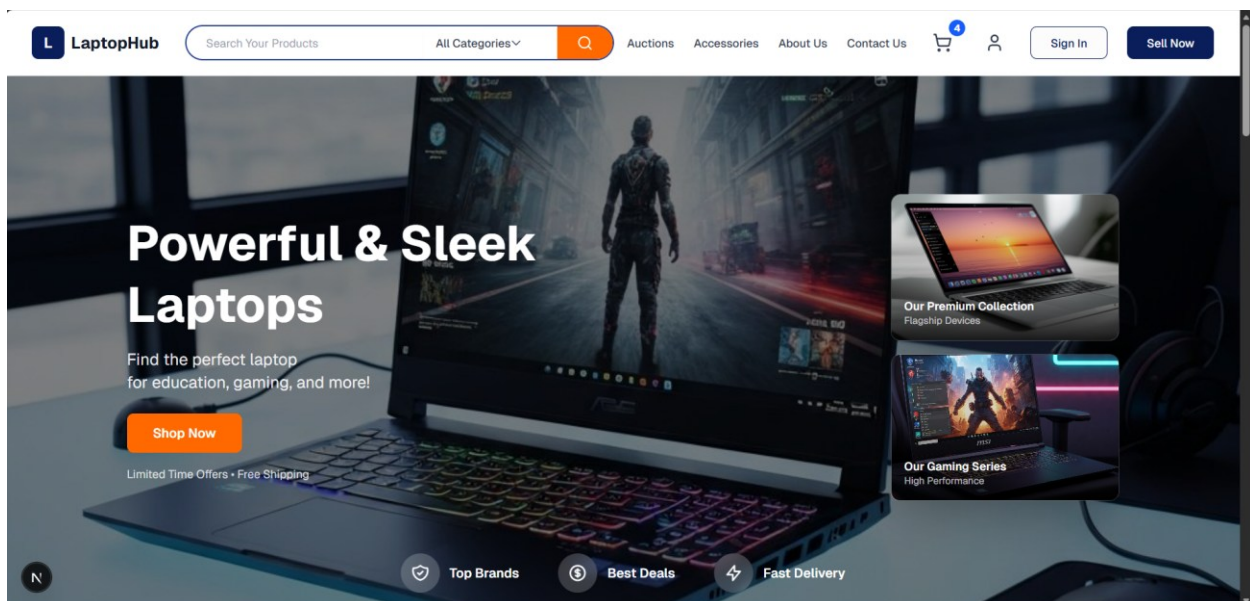


Figure 21 - Footer wireframe

3.2.2 System UI

The following images contain how the user interface will appear after development of the system.

- Home page



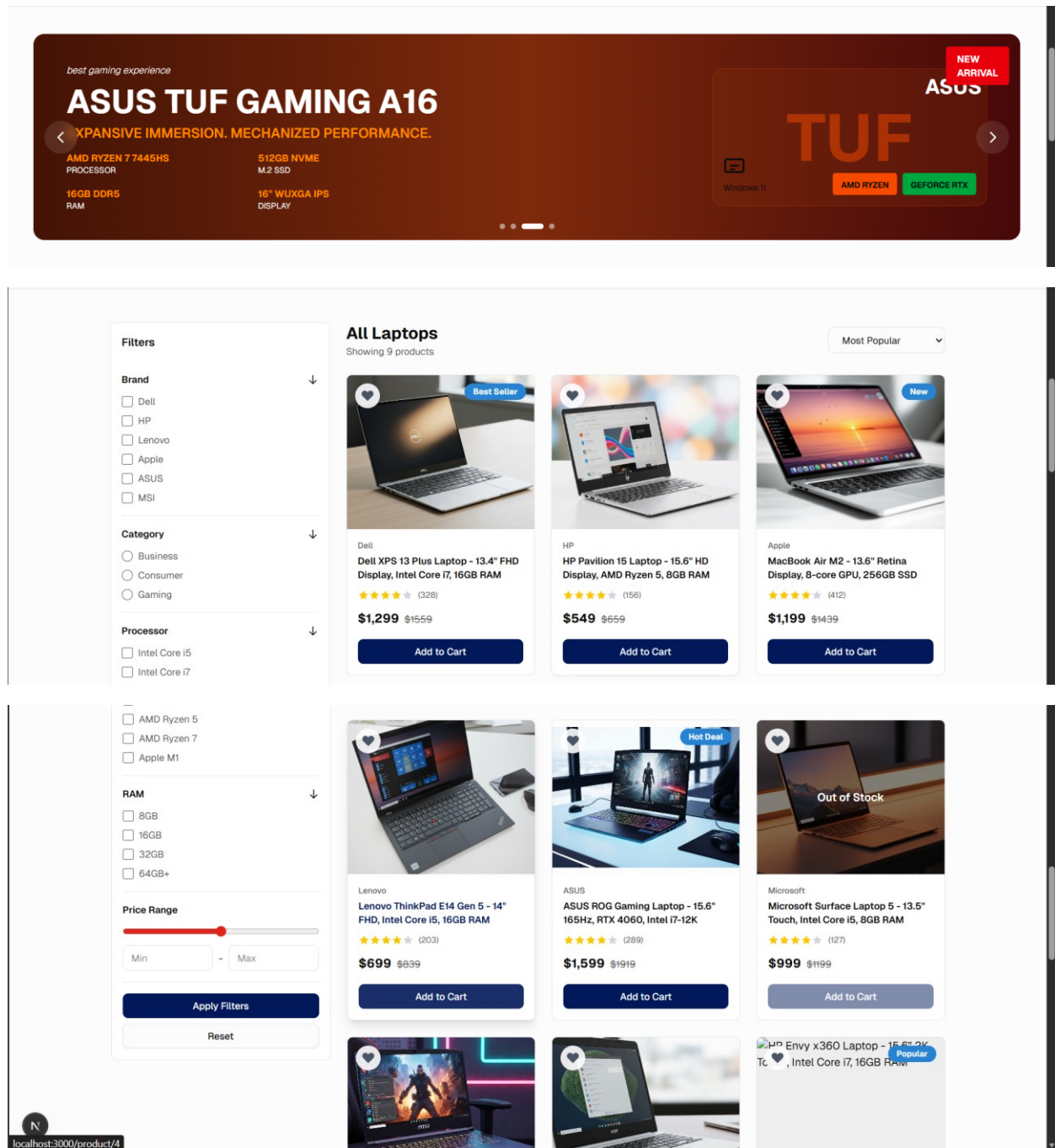


Figure 22 - Home page UI

- Auction page

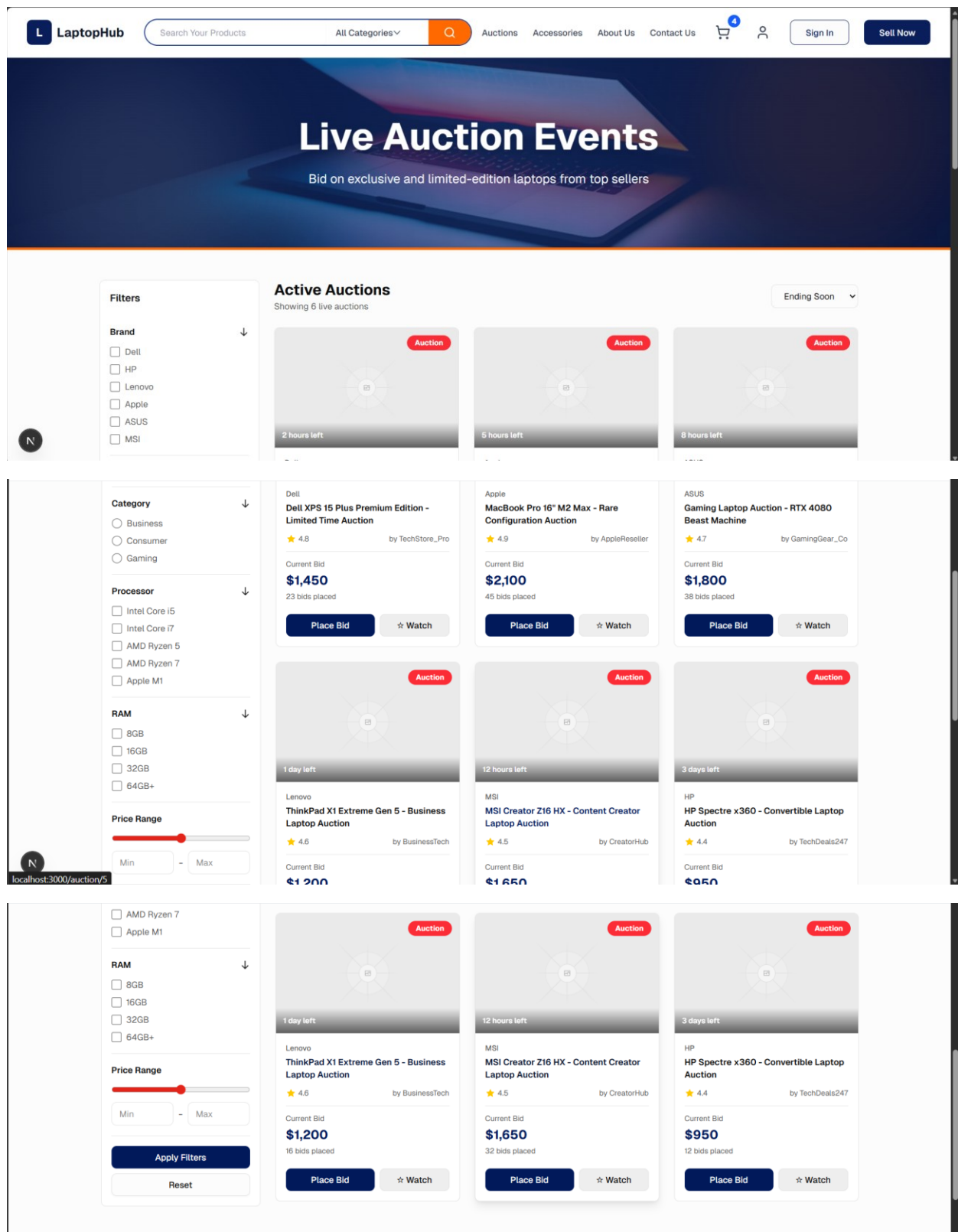


Figure 23 - Auction page UI

- Bid Confirmation page

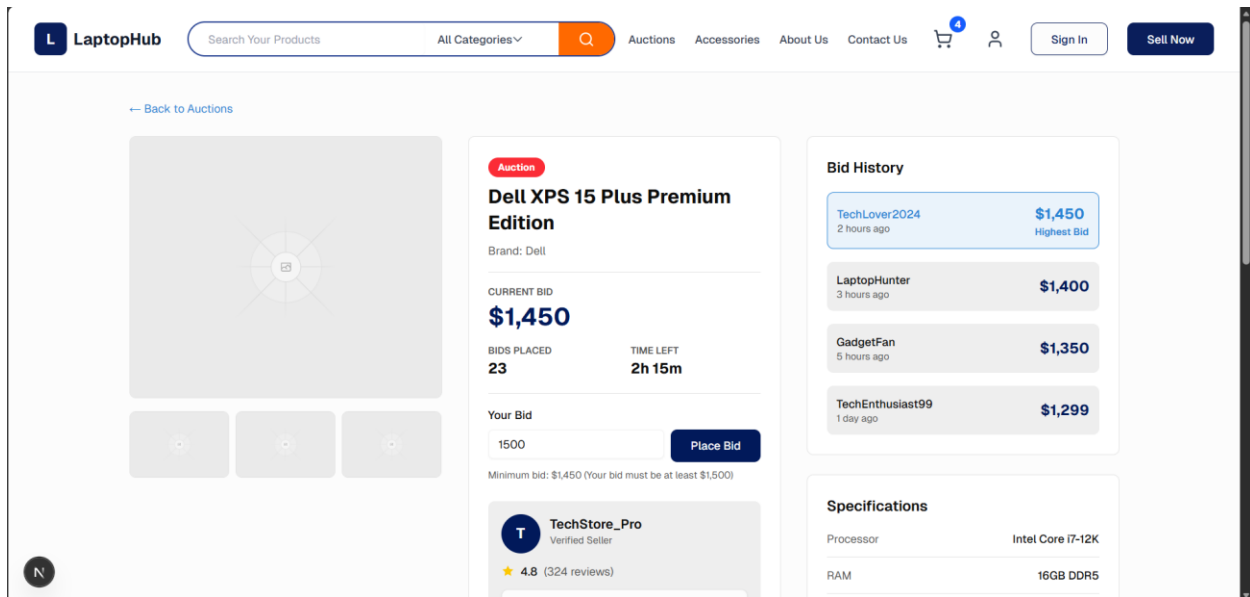
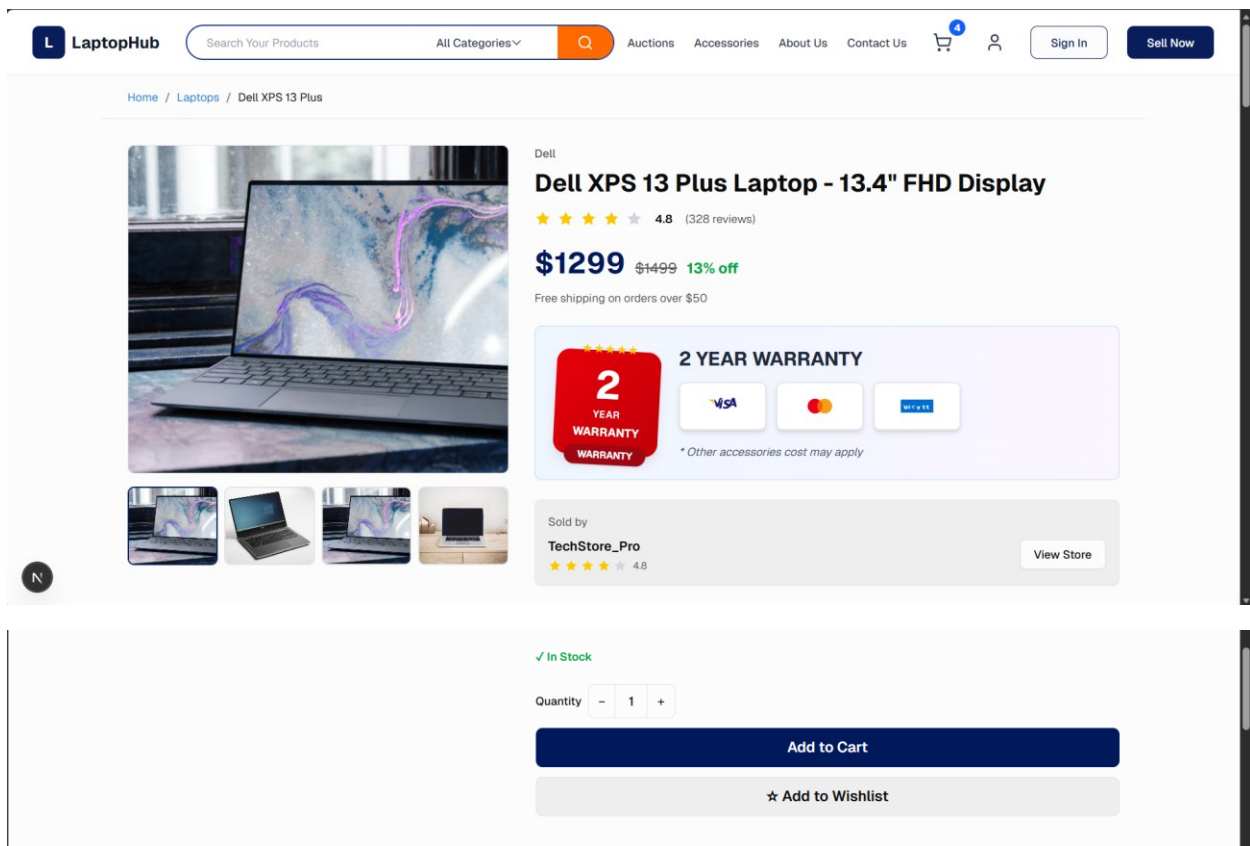


Figure 24 - Bid confirmation page

- Product page



7.

7.



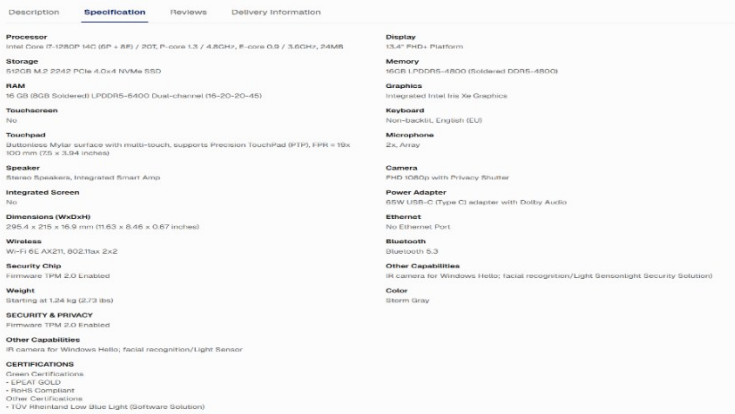
7.

7.



7.

2.



Find It Fast

- [Home](#)
- [Auctions](#)
- [About Us](#)
- [Our Products](#)
- [Contact Us](#)

Accessories

- Pendrive
- Headset
- Bags & Cases
- Mouse & Keyboards
- Virusguards
- Mouse Mat
- External HDD

- Pendrive
- Headset
- Bags & Cases
- Mouse & Keyboards
- Virusguards
- Mouse Mat
- External HDD

Technical Support:
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Technical Support:
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33

- Shopping cart

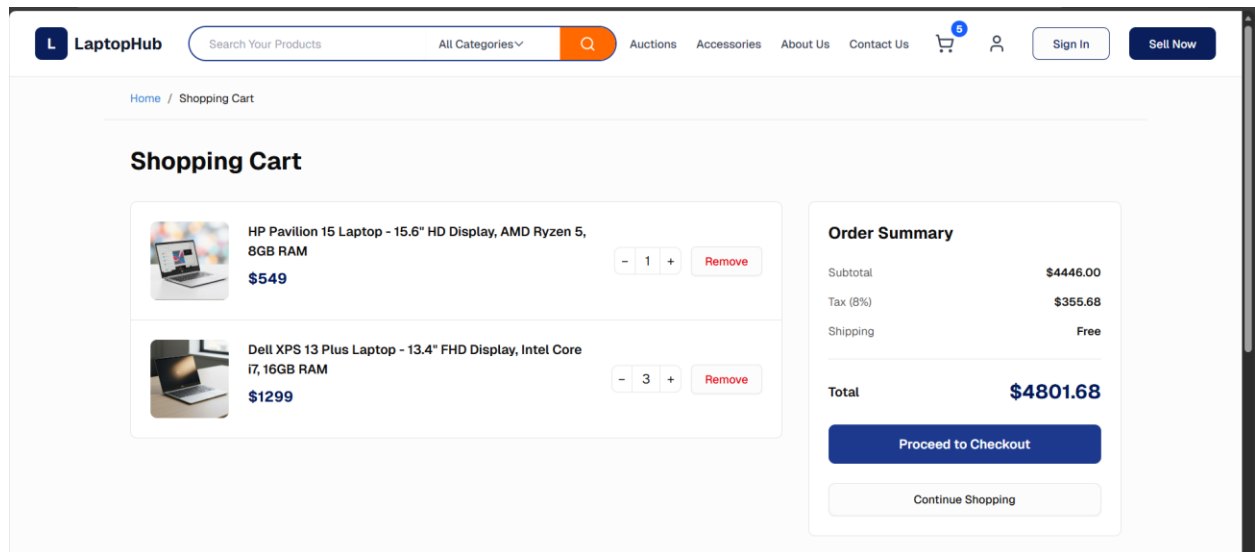
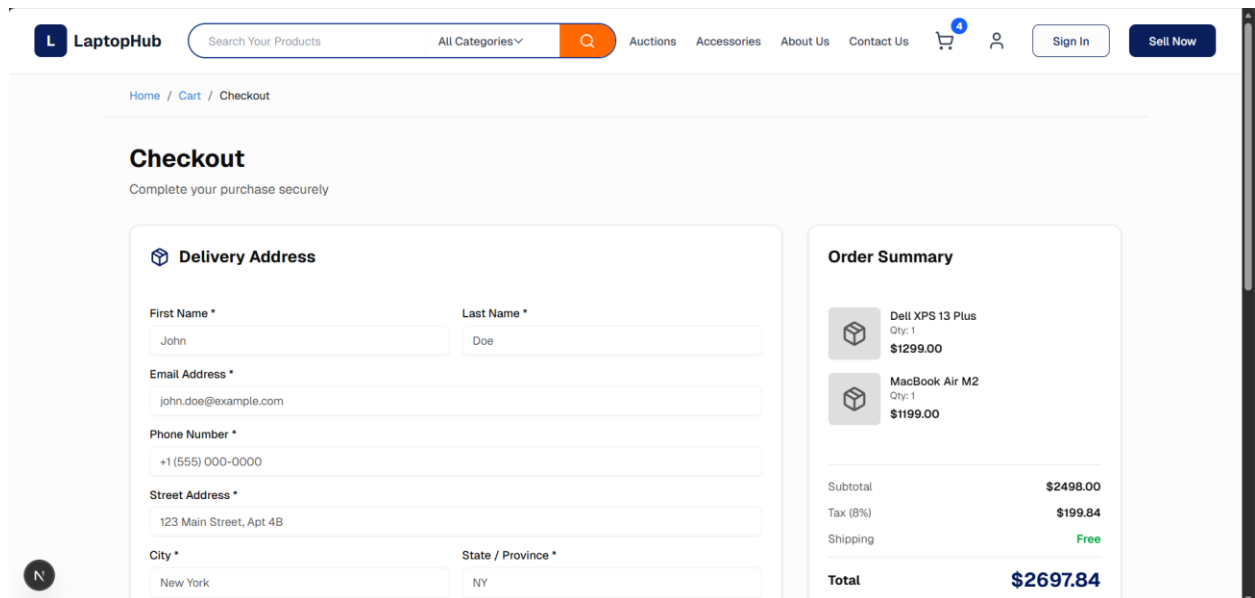


Figure 26 - Shopping cart UI

- Checkout Page



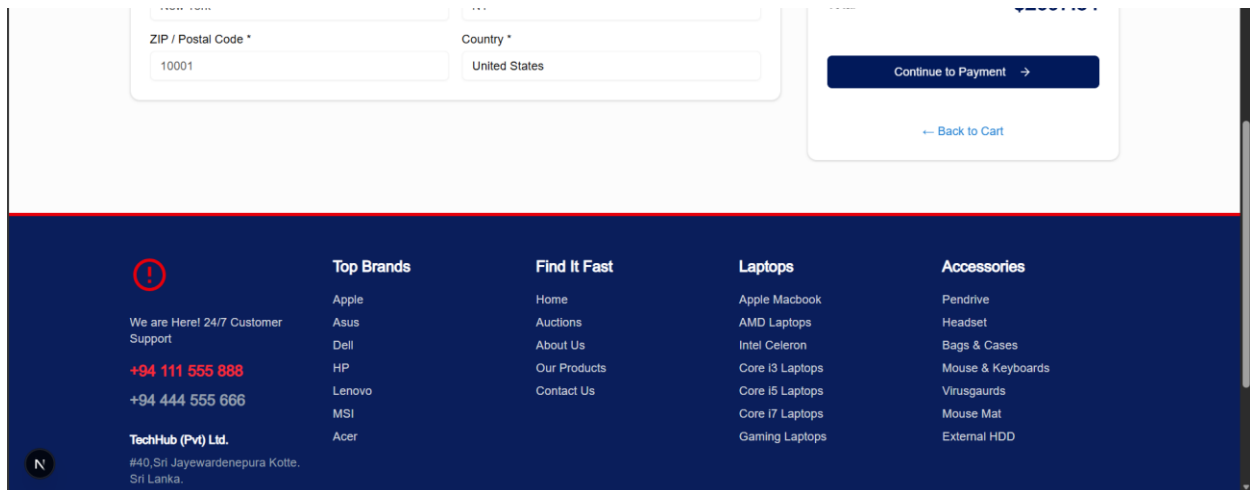


Figure 27 - Checkout page UI

• Payment Page

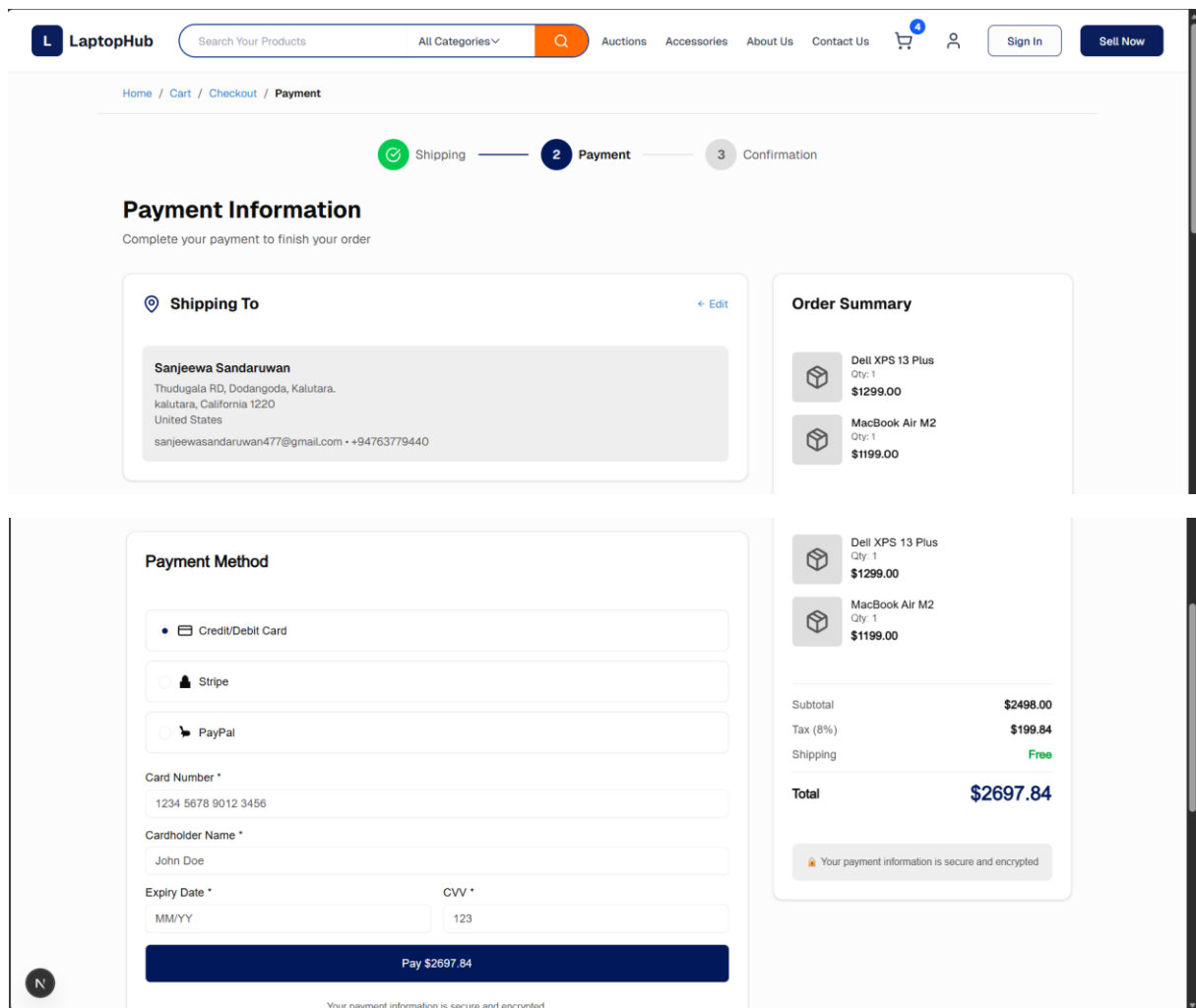


Figure 28 - Payment page UI

- Order Confirmation Page

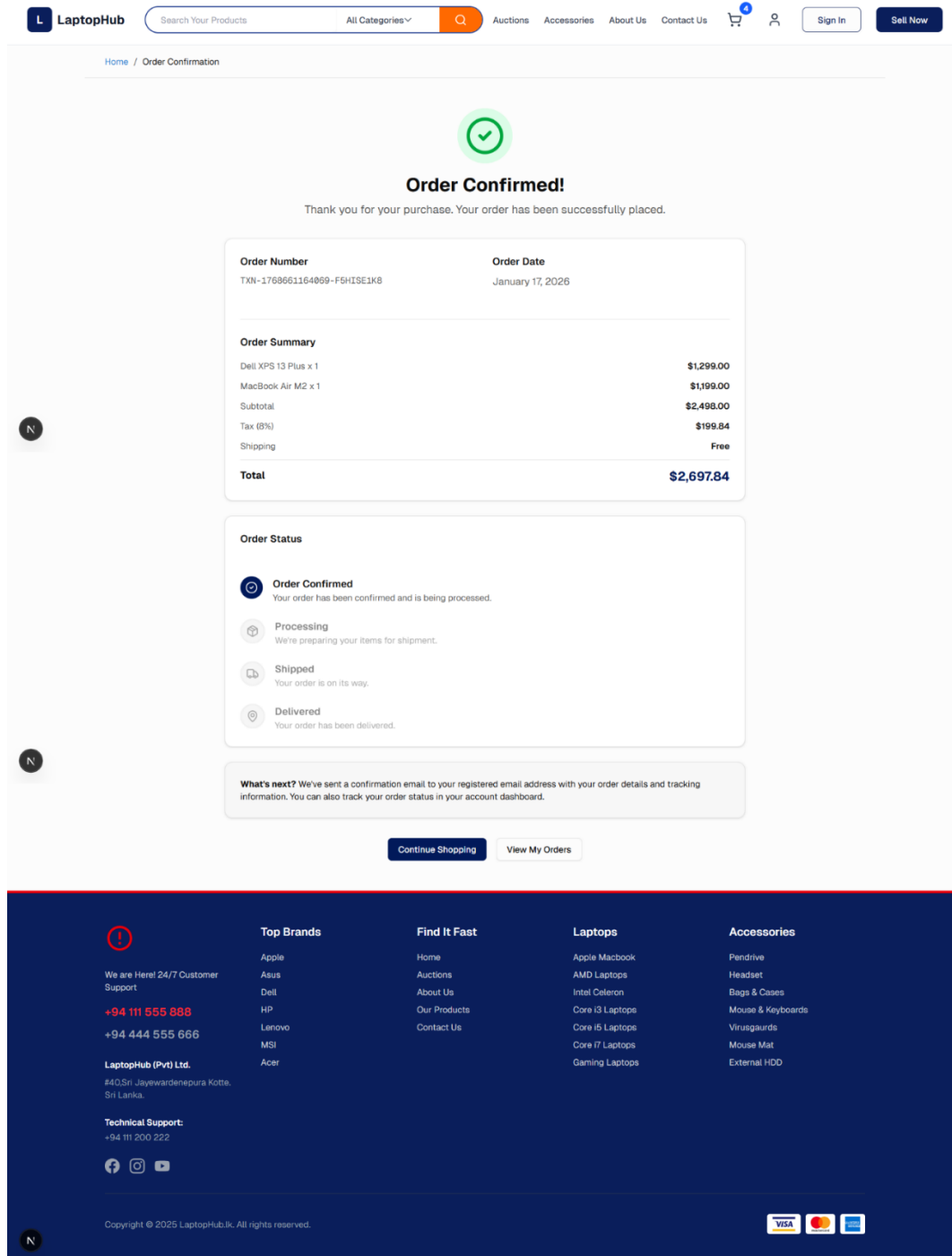


Figure 29 - Order confirmation page UI

- Computer accessories page

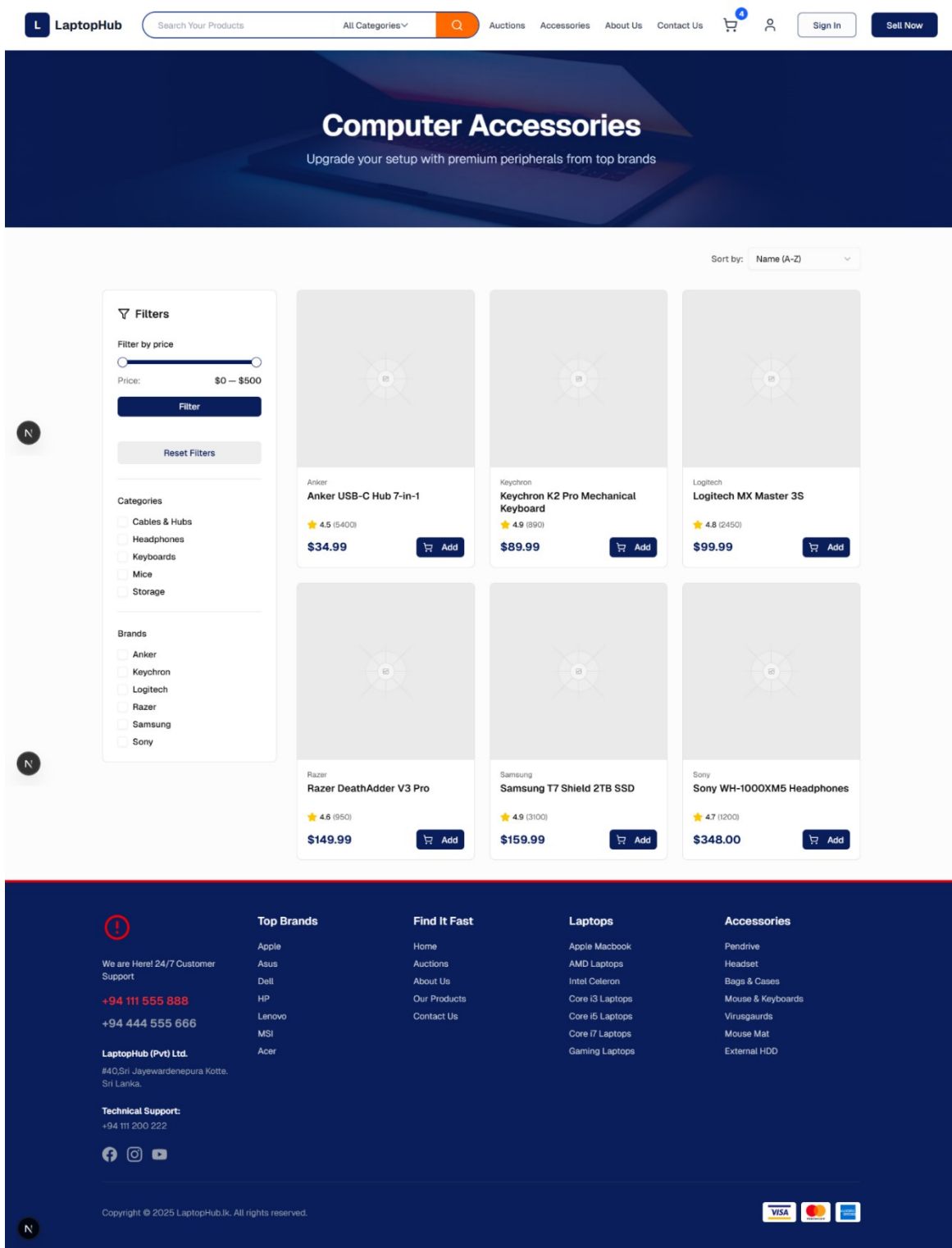


Figure 30 - Computer accessory page UI

- Account Page

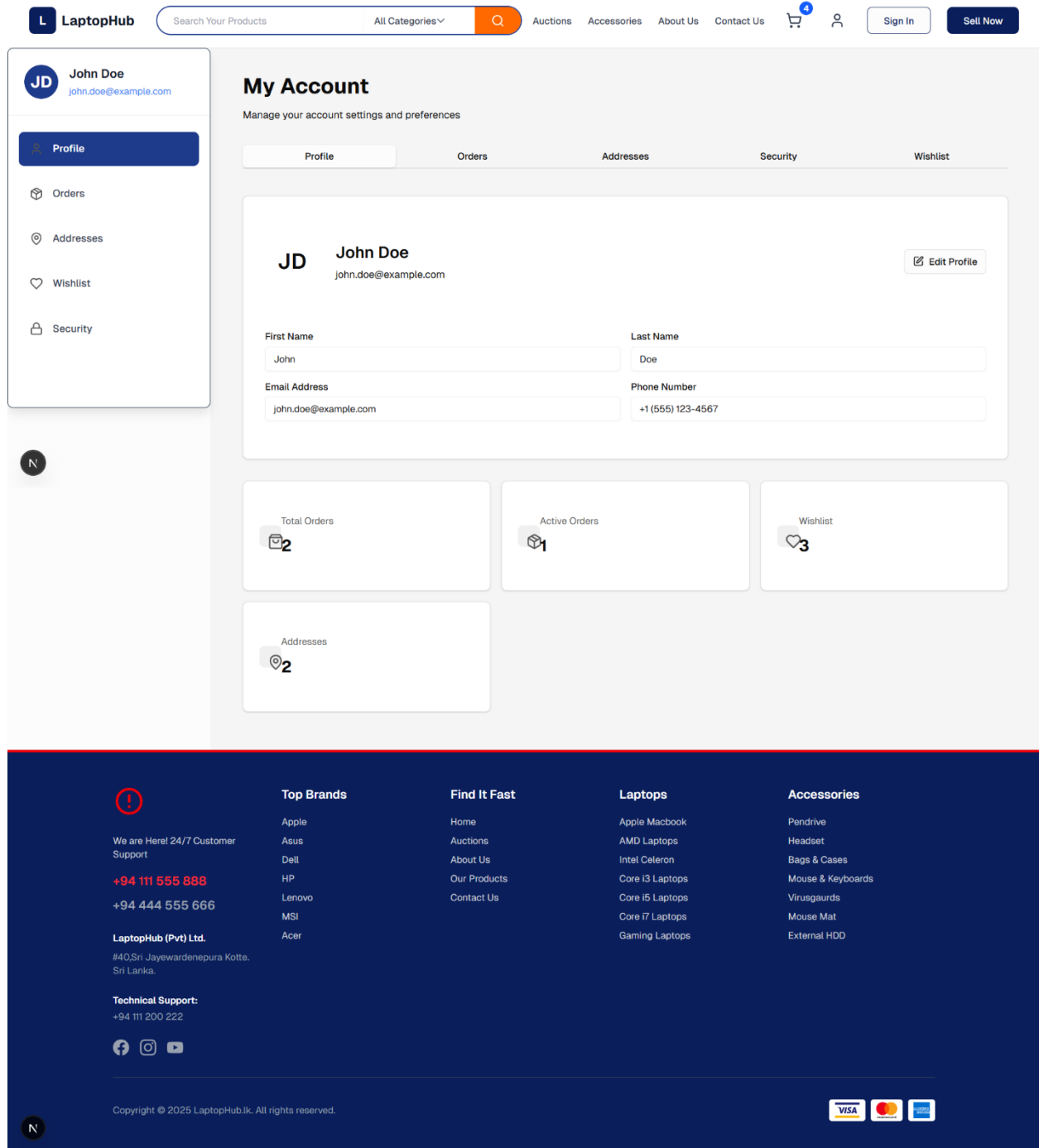


Figure 31 - Account page UI

- Admin dashboard

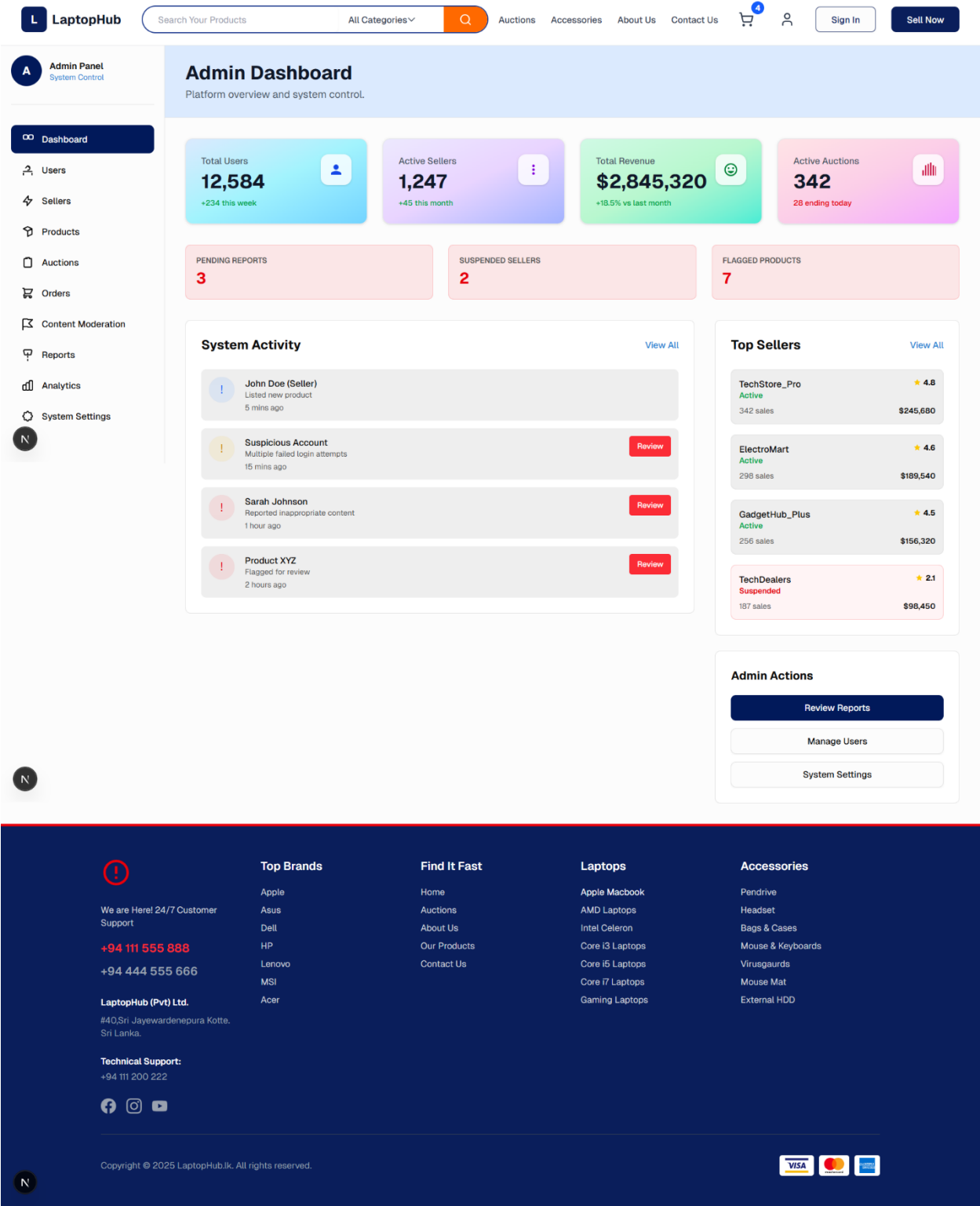


Figure 32 - Admin dashboard UI

- User Management page

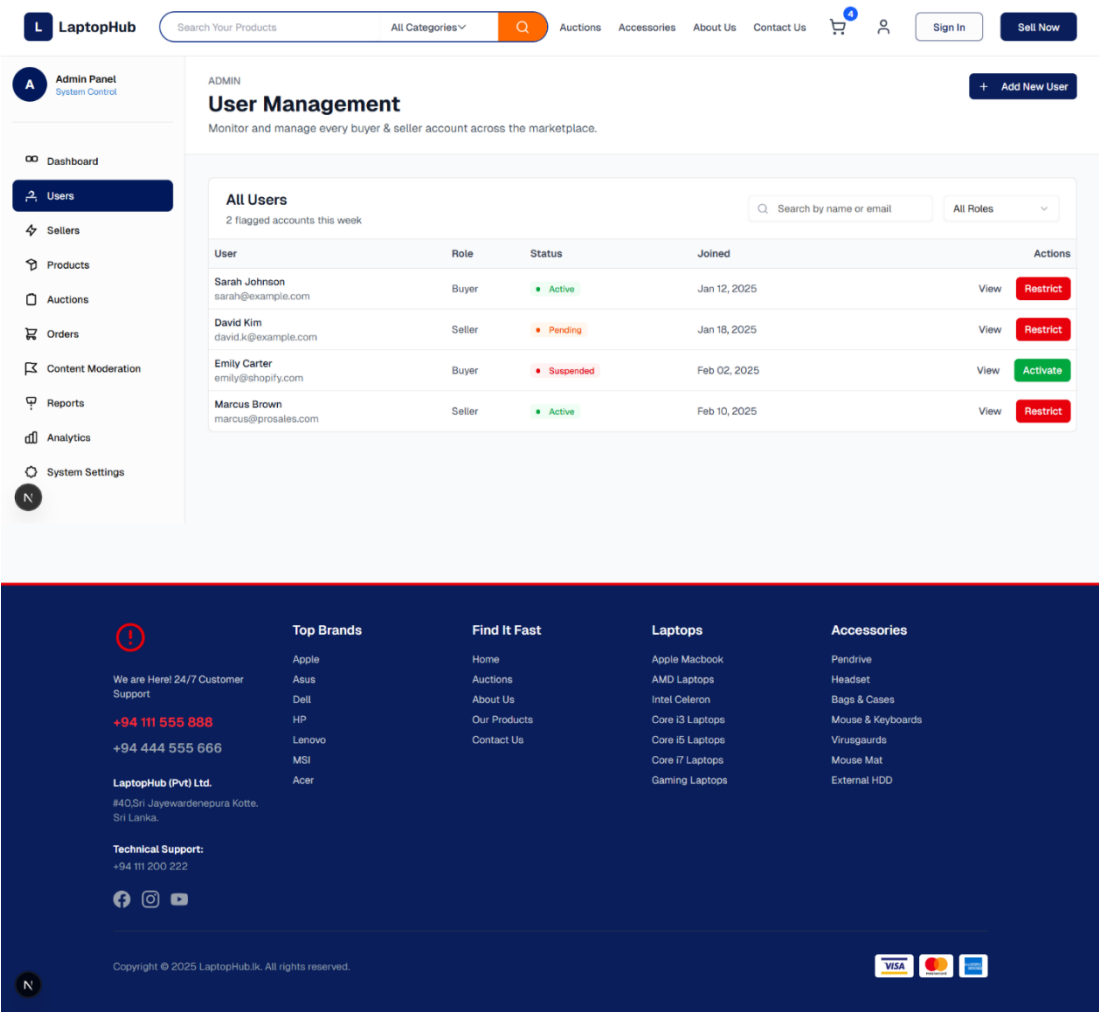


Figure 33 - User management page UI

- Sellers page

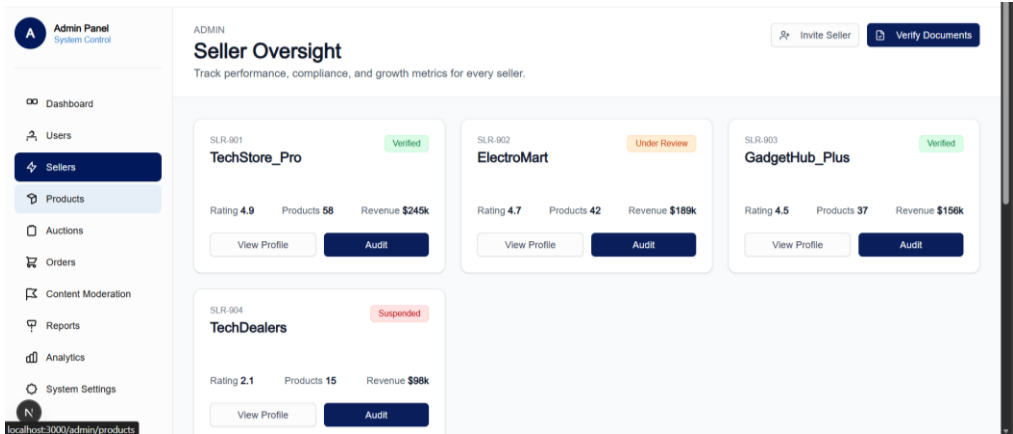


Figure 34 - Seller page UI

- Admin view - Products

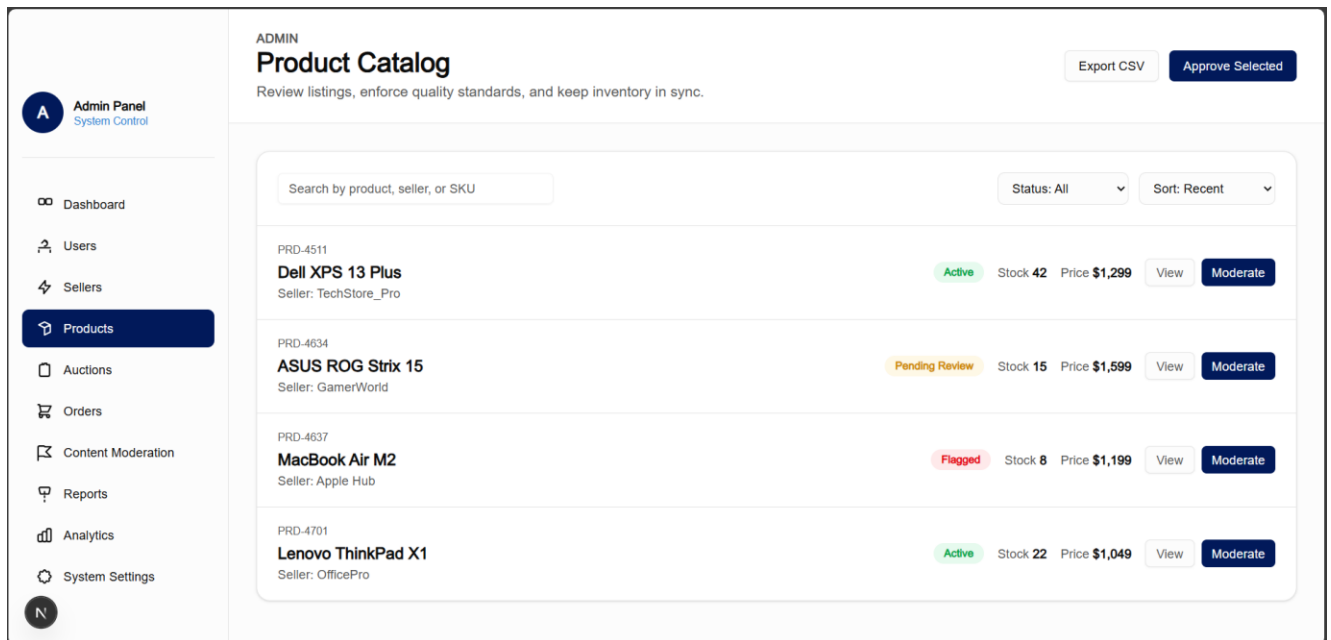


Figure 35 - Admin view - Products page UI

- Admin view - Auctions

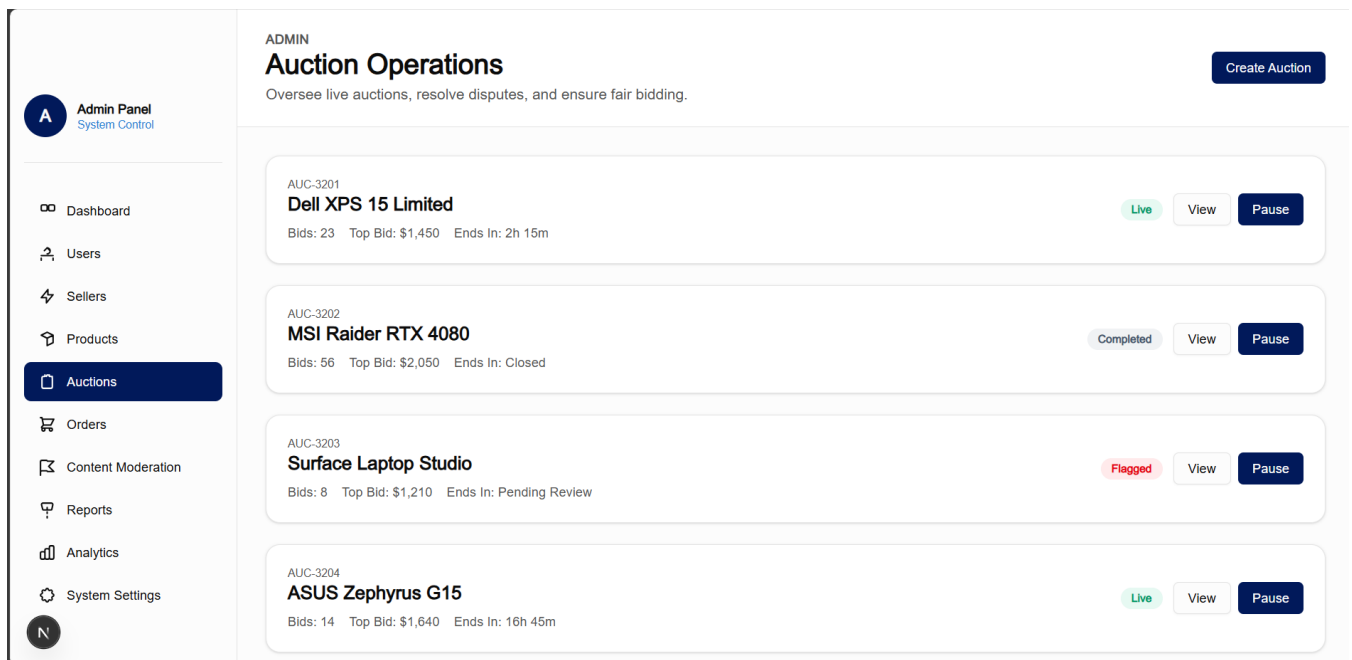


Figure 36 - Admin view - Auctions page UI

- Admin view - Orders

ADMIN
Order Pipeline
Track fulfillment, resolve disputes, and keep buyers informed.

[Generate Manifest](#)

Order	Customer	Method	Total	Status	Date	Actions
#ORD-9821	Sarah Johnson	Auction Win	\$1,450	Processing	Mar 12, 2025	Details Update
#ORD-9822	David Kim	Buy Now	\$2,050	Fulfilled	Mar 11, 2025	Details Update
#ORD-9823	Emily Carter	Auction Win	\$899	Dispute	Mar 11, 2025	Details Update
#ORD-9824	Marcus Brown	Buy Now	\$1,199	Pending Payment	Mar 10, 2025	Details Update

Figure 37 - Admin view - Orders page UI

- Admin view - Reports page

ADMIN
Executive Reports
Export-ready summaries for finance, trust & safety, and operations.

[Download PDF](#) [Schedule Report](#)

TOTAL REVENUE
\$2.84M
 +18.5% vs last month

ACTIVE LISTINGS
1,245
 +45 this month

DISPUTES
32
 -6 vs last week

SUSPENDED ACCOUNTS
12
 +3 urgent

Audit Trail
Recent actions that impact compliance [View All](#)

LOG-1290 Seller Verification TechStore_Pro	Completed Mar 12 • 09:24
LOG-1291 Manual Refund Order #9823	Pending Approval Mar 12 • 08:51
LOG-1292 Listing Flagged MacBook Air M2	Escalated Mar 11 • 22:34

localhost:3000/admin/dashboard

Figure 38 - Admin view - Reports page UI

- Analytics page

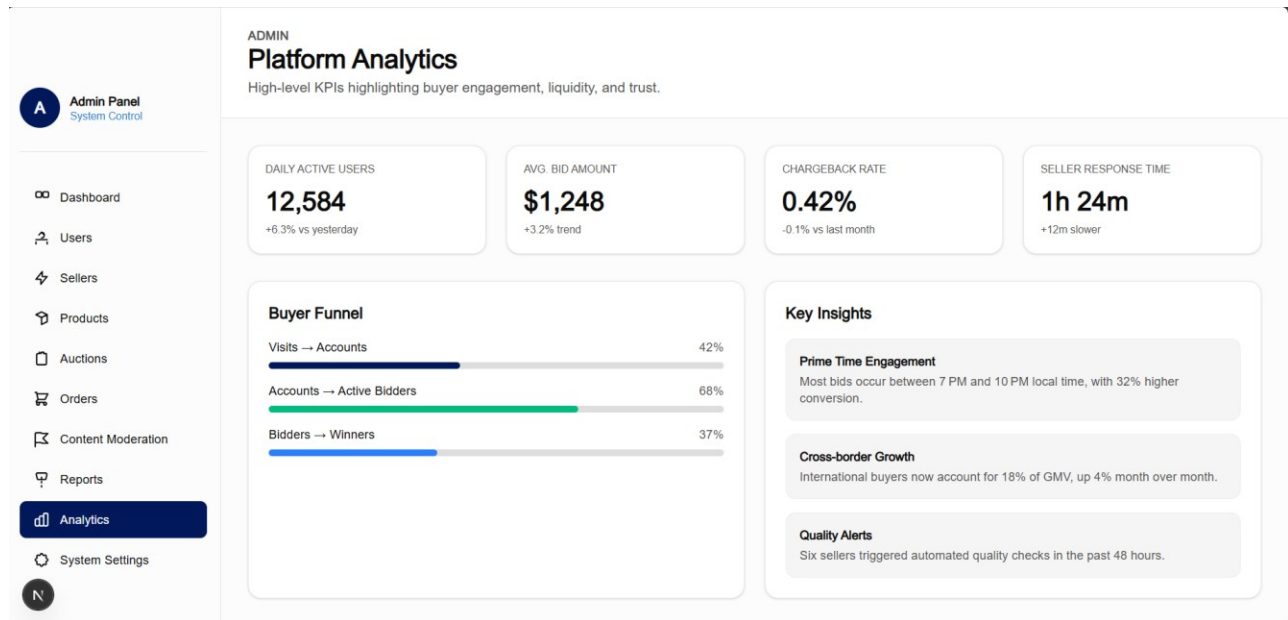


Figure 39 - Analytics page UI

- Seller Dashboard page

L LaptopHub

Search Your Products

All Categories▼

Q

[Auctions](#)
[Accessories](#)
[About Us](#)
[Contact Us](#)

4

Sign In

Sell Now

TS TechStore_Pro

Verified Seller

View Store

Seller Dashboard

Welcome back! Here's your sales overview.

Dashboard

My Products

Add Product

My Auctions

Orders

Messages

Analytics

Store Settings

Total Sales

\$124,568

+12.5% vs last month

Orders

234

+8 orders today

Products

48

5 new this week

Customer Rating

4.8

Based on 328 reviews

Recent Orders

View All

Order ID	Product	Customer	Amount	Status	Date
ORD001	Dell XPS 13	John Doe	\$1,299	Delivered	2 hours ago
ORD002	MacBook Air M2	Jane Smith	\$1,199	Shipped	5 hours ago
ORD003	Lenovo ThinkPad	Bob Wilson	\$699	Processing	1 day ago
ORD004	HP Pavilion	Alice Brown	\$549	Pending	2 days ago

Top Products

Dell XPS 13 Plus

42 sales

\$54,558

+12%

MacBook Air M2

28 sales

\$33,572

+8%

Lenovo ThinkPad E14

19 sales

\$13,281

+5%

Quick Actions

Add New Product

Create Auction

View Messages

! We are Here! 24/7 Customer Support

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+94 444 555 666

LaptopHub (Pvt) Ltd.

#40,Sri Jayewardanepura Kotte, Sri Lanka.

Technical Support:

+94 111 200 222

Top Brands

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[Asus](#)
[Dell](#)
[HP](#)
[Lenovo](#)
[MSI](#)
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[AMD Laptops](#)
[Intel Celeron](#)
[Core i3 Laptops](#)
[Core i5 Laptops](#)
[Core i7 Laptops](#)
[Gaming Laptops](#)

Accessories

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[Headset](#)
[Bags & Cases](#)
[Mouse & Keyboards](#)
[Virusguards](#)
[Mouse Mat](#)
[External HDD](#)

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Figure 40 - Seller dashboard UI

- Seller view - My products page

Product	Price	Stock	Status	Rating	Sales	Actions
 Dell XPS 13 Laptop ID: 1	\$999	12 units	Active	4.8 ★ (234)	156	  
 HP Pavilion 15 ID: 2	\$649	8 units	Active	4.5 ★ (156)	89	  
 MacBook Air M2 ID: 3	\$1299	0 units	Inactive	4.9 ★ (512)	234	  
 Lenovo ThinkPad ID: 4	\$849	15 units	Active	4.6 ★ (289)	145	  
 ASUS ROG Gaming Laptop ID: 5	\$1599	5 units	Draft	4.7 ★ (178)	92	  
 Microsoft Surface Laptop ID: 6	\$1199	10 units	Active	4.4 ★ (267)	178	  

Figure 41 - Seller view - My products page UI

- Seller view - add products page

Add New Product

Create a new product listing for your store

Basic Information

Product Name
e.g., Dell XPS 13 Laptop

Category
Laptops

Description
Describe your product in detail...

Pricing & Stock

Price (\$)
999.99

Stock Quantity
10

Product Images

Drop images here or click to upload
PNG, JPG up to 10MB each

Tips for Better Listings

- Use clear, descriptive product names
- Include detailed specifications
- Add high-quality product images
- Set competitive pricing
- Provide accurate stock information

Figure 42 - Seller view - add products page UI

- Seller view - My auctions page

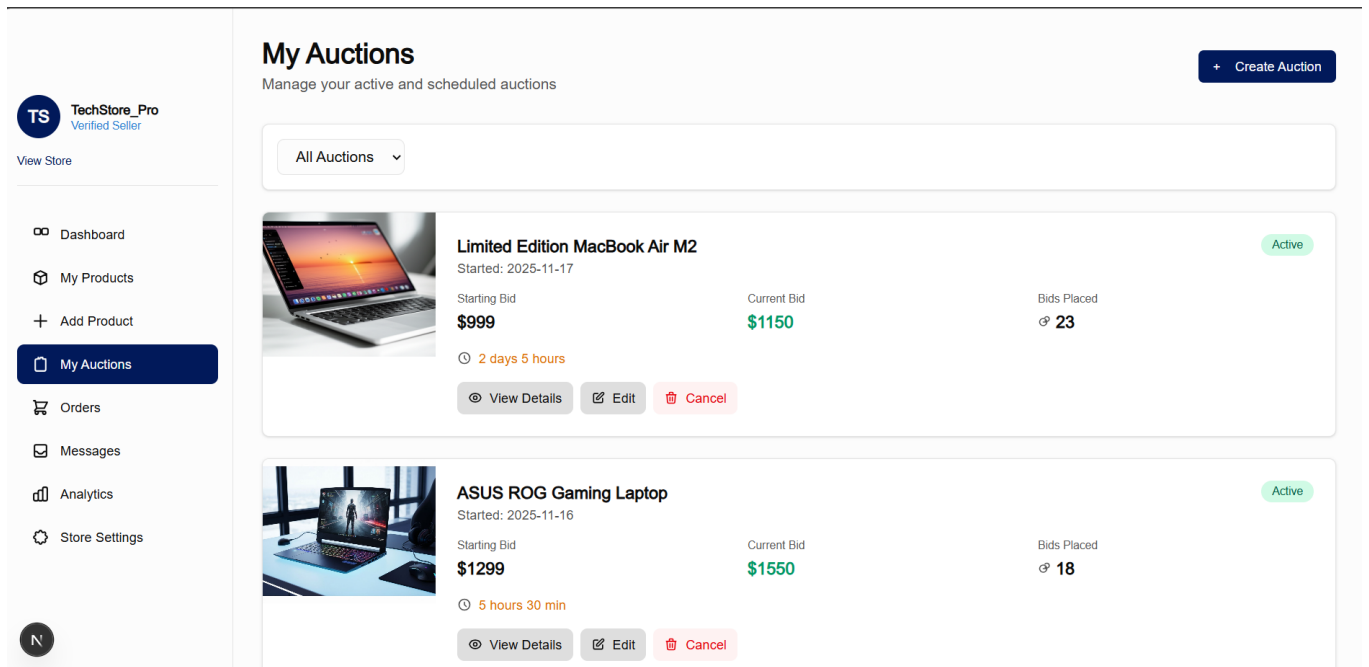


Figure 43 - Seller view - My auctions page UI

- Seller view - Orders page

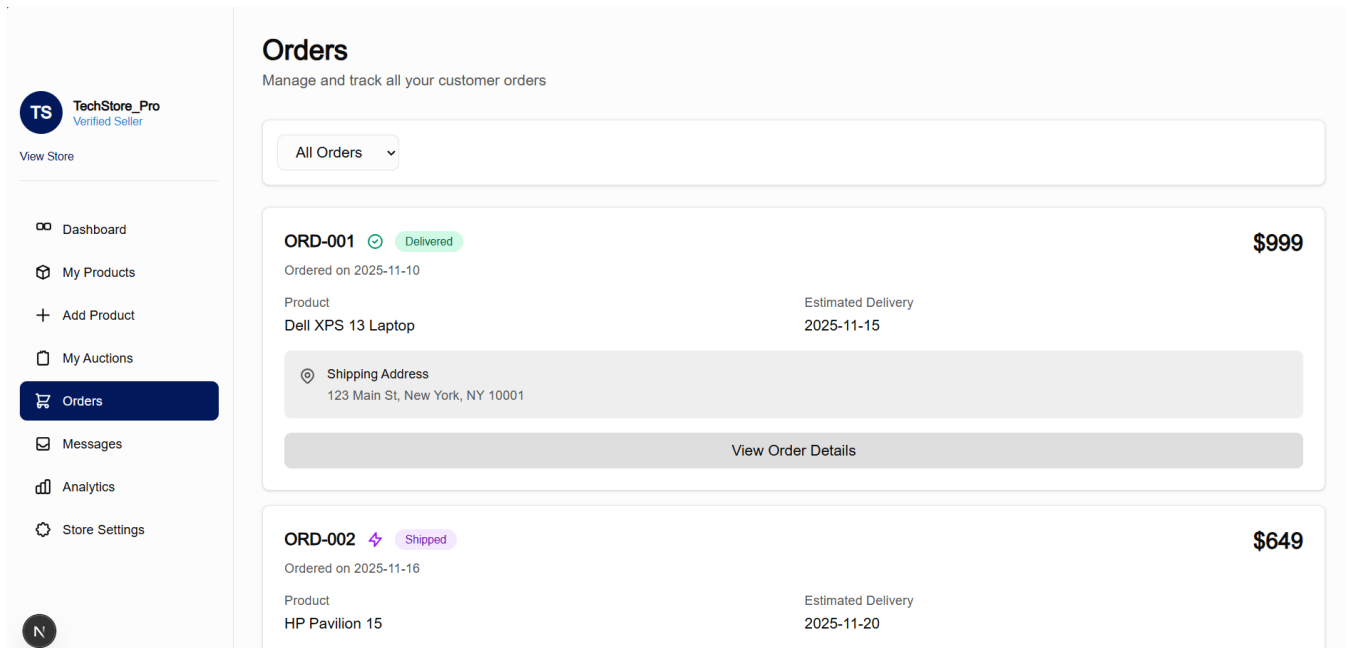


Figure 44 - Seller view - Orders page UI

- Seller view - Analytics page

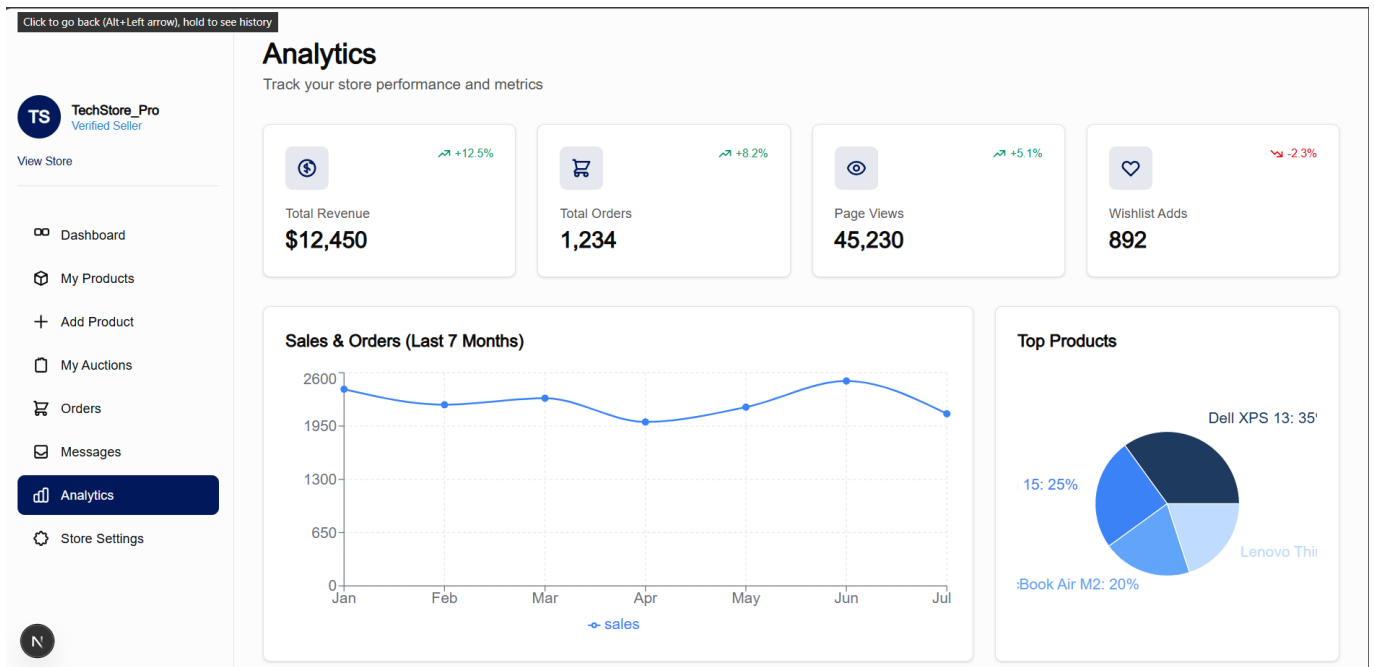


Figure 45 - Seller view - Analytics page UI

- Sign In page

LaptopHub

The best place to buy and sell premium laptops. Secure, fast, and reliable.

Welcome back

Enter your email to sign in to your account

Email Address
admin@gmail.com

Password

☐ Remember me [Forgot password?](#)

[Sign In](#)

Or continue with

[Google](#) [Facebook](#)

Don't have an account? [Sign up](#)

Figure 46 - Sign in page UI

- Sign up page

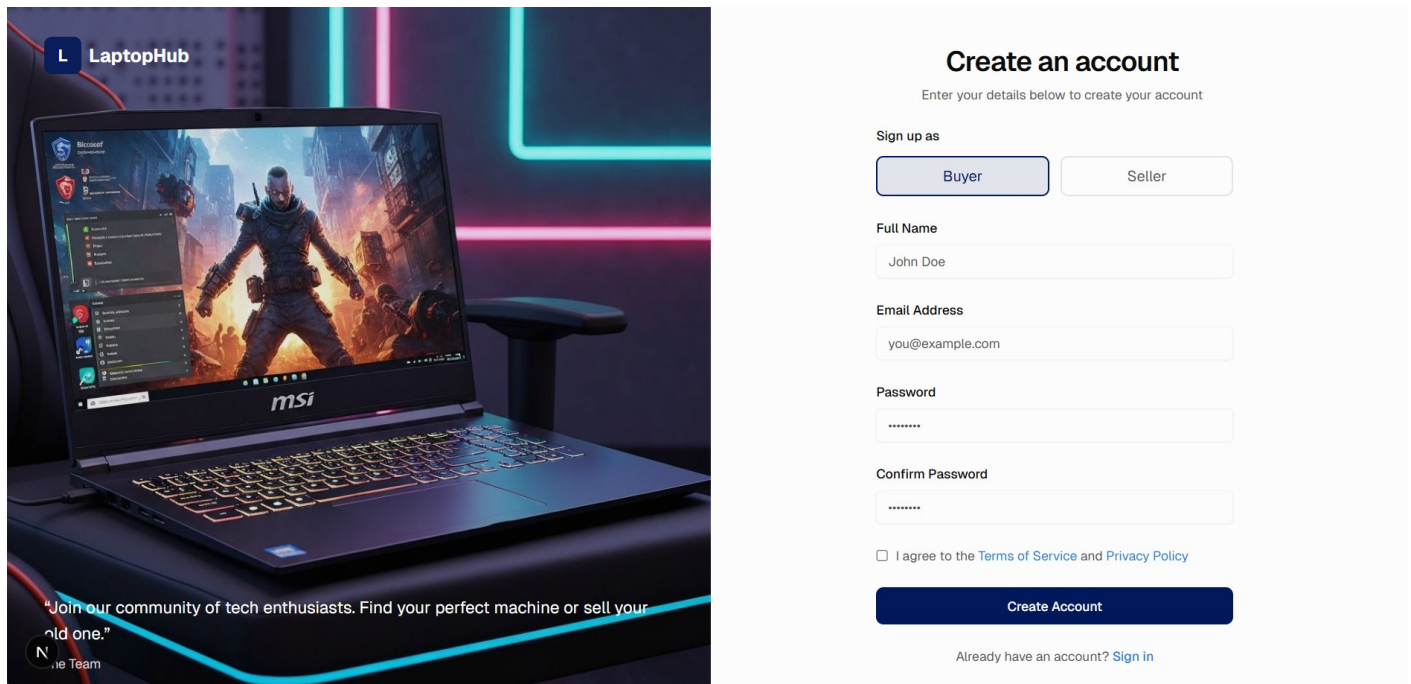


Figure 47 - Sign up page UI

4. Conclusion

The proposed Centralized E-Commerce Platform for Laptop Sales and Auction Management aim to provide an efficient, user-friendly platform for both customers and sellers in the electronic devices market. By integrating product browsing, comparison, direct purchasing and auction-based selling within a single platform, this system chooses to address a specific challenge often faced by buyers in selecting a quality product, especially a laptop. Furthermore, this system allows sellers to reach a wider customer base.

The platform ensures secure access control and role-based functions through clearly defined roles within the system. Through automation of main functionalities such as inventory management, order processing, payment handling and customer data management, the system automates many manual effort instances and other operational errors encountered within manual systems.

This project aims to deliver a scalable, centralized web-based solution that enhances the buying and selling experience of laptops and other electronic products while providing support for auction of the said products. This platform enables sellers from all scales, from individual sellers to recognized major sellers, to participate in this effective digital marketplace.

5. Team details

Register No	Name	Role
FC115278	M.A.O. Rodrigo	Project manager
FC116345	U.J.N.S. Tharaka	Analyst
FC115733	S. Gunaruban	Designer
FC115528	D.P. Gamage	Database administrator
FC115659	U.S. Ravihara	Full stack developer
FC115250	U.K.D.S. Sandaruwan	Front-end developer
FC115558	L.R. Thuduwege	Back-end developer
FC115592	K.G.A.K. Sathsarani	Quality assurance engineer

Table 4 - Team details